

Chapter 4

Moments and Static Equivalence

When a force acts on a body, it can potentially produce two effects: translation of the body in the direction of the force, and rotation of the body about an axis. For a body in equilibrium, we say that forces have a *tendency* to produce translation or rotation since no actual acceleration or motion occurs.

The rotational tendency of a force is the subject of this chapter. Engineers refer to this rotational tendency as a **moment**, short for “moment of a force.” You may remember using the term **torque** for the same quantity in physics. Physicists use *torque* specifically for the moment of a force, and *moment* for the product of *any* physical quantity with distance. To an engineer, a torque refers specifically to a moment about the long axis of an object that produces twisting and torsional stresses.

A wrench provides a familiar example. A force $\mathbf{F}$ applied to the handle of a wrench, as shown in [Figure 4.0.1](#), creates a moment $\mathbf{M}_A$ about an axis perpendicular to the page through the center of the nut at point A . The $\mathbf{M}$ is bold because moments are vector quantities, and the subscript A indicates the axis or center of rotation. The direction of the moment can be either clockwise or counterclockwise, depending on how the force is applied. If the nut is frozen, no actual rotation occurs, but the force still produces a tendency to rotate it, i.e. a moment.

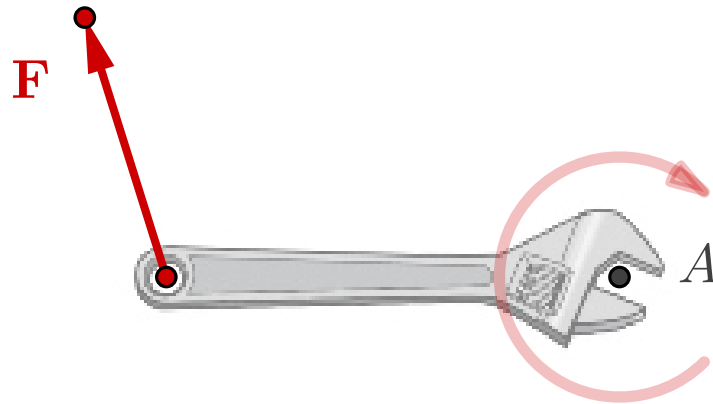
[Live Link](#)

Figure 4.0.1 A moment M_A is created about point A by force $\mathbf{F}$.

4.1 Moment of Force

A moment of force, or torque, is a measure of the tendency of that force to rotate a body about a selected point or axis, called the **moment center**. This tendency increases with the magnitude of the force, and also with the distance between the line of action of the force and the moment center.

Moments are vector quantities, so they have magnitude and direction and obey all the rules of vector arithmetic, even dot and cross products, described in [Chapter 2](#).

4.1.1 Magnitude of a Moment

The magnitude of a moment describes how hard it turns, in the same way that the magnitude of a force describes how hard it pushes or pulls. When you turn a door knob, the magnitude of the moment is how hard you twist the knob.

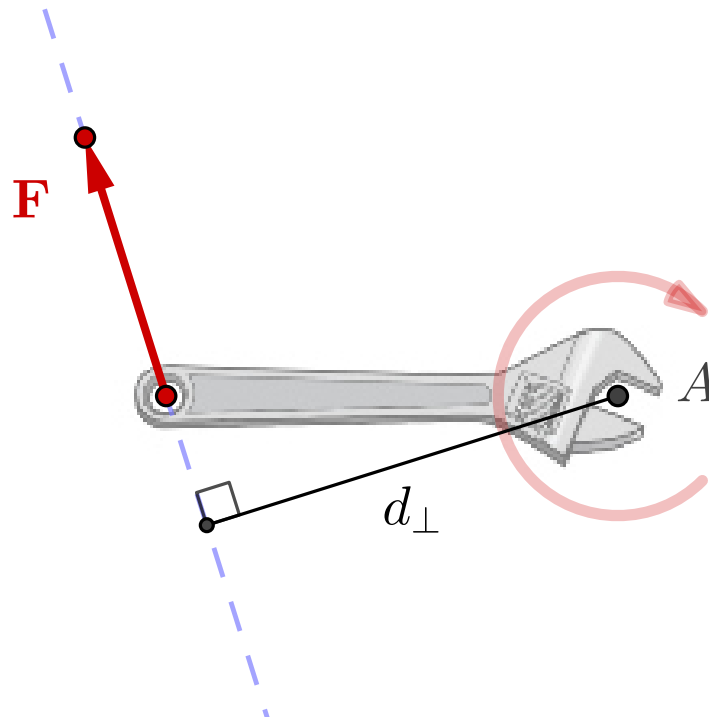
The magnitude of a moment is found by multiplying the magnitude of the force by the distance between the line of action of the force and the center of

rotation, as illustrated in Figure 4.1.1. This distance called the **moment arm** or the **perpendicular distance**, $d_{\perp}$.

$$M = Fd_{\perp}. \quad (4.1.1)$$

Since the moment is the product of the force's magnitude and the perpendicular distance, the closer the point is to the force's line of action, the smaller the moment. If the point lies upon the force's line of action, then the moment arm is zero, making the moment zero as well. But notice that a force may be slid along its line-of-action without changing the moment, because neither the magnitude of the force nor the moment arm changes. The magnitude of a moment is a positive quantity regardless of whether it produces a clockwise or counterclockwise tendency.

Moments are the product of a force with a distance, so they have units of [force] $\times$ [distance] such as N-m or ft-lb.



[Live Link](#)

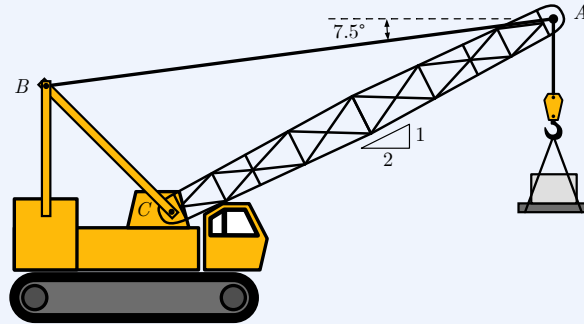
Figure 4.1.1 Definition of the moment, $M = Fd_{\perp}$.

Example 4.1.2

A mobile crane is used to raise a 500 lb load of building materials. Boom AC is 40 ft long and the tension of the boom pendant AB is 1500 lb.

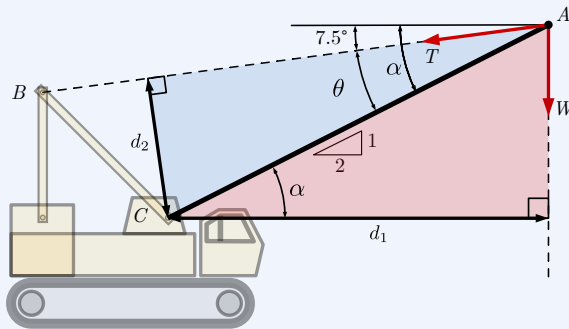
Use the definition of the moment (4.1.1) to determine the magnitude and direction of:

- M_1 , the moment about point C caused by the load, W .
- M_2 , the moment about point C caused by tension T .



Solution.

- Given: $W = 500$ lb, $T = 1500$ lb, boom length $AC = 40$ ft.
- Sketch right triangles containing the perpendicular distances d_1 and d_2 from point C to the lines of action of W and T .



- Find angles in the right triangles.

$$\alpha = \tan^{-1}(1/2) = 26.6^\circ$$

$$\theta = \alpha - 7.5^\circ = 19.1^\circ$$

- Use trigonometry to find the perpendicular distances.

$$d_1 = 40 \cos \alpha = 35.8 \text{ ft}$$

$$d_2 = 40 \sin \theta = 13.06 \text{ ft}$$

- Apply the definition of the moment (4.1.1), and solve for the moments.

$$M_1 = Wd_1 = (500 \text{ lb})(35.8 \text{ ft}) = 17,885 \text{ ft}\cdot\text{lb}$$

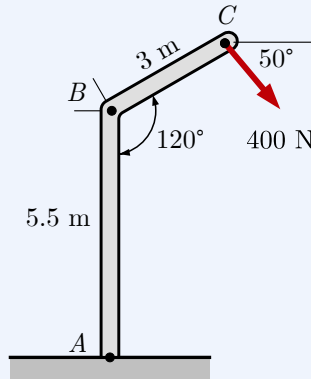
$$M_2 = Td_2 = (1500 \text{ lb})(13.06 \text{ ft}) = 19,590 \text{ ft}\cdot\text{lb}$$

By inspection, we see that the tension produces a counterclockwise rotation and the load produces a clockwise rotation. Counterclockwise moment M_2 is greater than the clockwise moment M_1 because the boom pendant must also support the weight of the boom, which we have not accounted for.

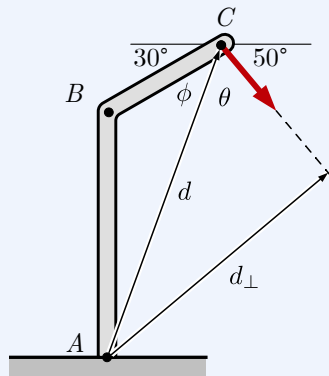
As we will see in the next section, counterclockwise moments indicate a positive scalar moment in two dimensions, and a vector that points in the $+\mathbf{k}$ direction in three.

Example 4.1.3

The bent column ABC is firmly fixed to the ground at A . Determine the magnitude and direction of the moment about point A produced by the 400 N force applied at C .



Solution. This problem is similar to the previous example, but the geometry required to find the perpendicular distance is slightly more difficult.



1. Use the law of sines and law of cosines to find distance d from A to C and the angle ϕ in triangle ABC .

$$d^2 = 5.5^2 + 3^2 - 2(3)(5.5) \cos 120^\circ$$

$$d = 7.47 \text{ m}$$

$$\frac{d}{\sin 120^\circ} = \frac{5.5 \text{ m}}{\sin \phi} \implies \phi = 39.6^\circ$$

2. Then, find the angle θ in the right triangle and solve for the perpendicular distance.

$$\theta = 180^\circ - 30^\circ - 50^\circ - \phi = 60.36^\circ$$

$$d_\perp = d \sin \theta = 6.49 \text{ m}$$

3. Finally, use (4.1.1) to find the magnitude of M_A created by 400 N force F .

$$M_A = F d_\perp = 2596 \text{ N}\cdot\text{m}$$

This force produces a clockwise rotation of the column, so the corresponding scalar moment is negative and the moment expressed as a vector is

$$\mathbf{M}_A = 2596 \text{ N}\cdot\text{m} \odot$$

4.1.2 Direction of a Moment

In two-dimensional problems, moment vectors are perpendicular to the page and moments simplify to clockwise or counterclockwise rotation about a *point*, called the **point of interest** or **moment center**. The point of interest can be visualized as the point where the axis of rotation pierces the page. Since these moment vectors point into the third dimension, they are usually drawn as circular arrows in the plane of the page.

Consider the two moments acting on a horizontal surface, shown below. To determine the *sense* of these moments, i.e. whether the vectors point up or down, apply the right-hand rule. Imagine grasping the axis of rotation with your right hand, with your fingers curling in the direction of rotation; your thumb then points in the direction of the moment vector. Using this technique, we observe that counterclockwise moments point upward, or out of the page, while the clockwise moments point downward or into the page. In other words, counterclockwise moment vectors point towards the positive z direction and clockwise moments point in the $-z$ direction.

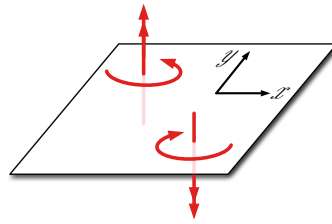


Figure 4.1.4 Moments in the plane of the page.

Pause your reading now to apply the right-hand rule yourself to determine the direction of the two moments in [Figure 4.1.4](#)

In three dimensions, moment vectors are more difficult to visualize, because they can point in any direction in 3-D space, however they still point along an axis of rotation with sense determined by the right hand rule.

As discussed in [Subsection 2.8.1](#), there are several alternative methods for applying the right-hand rule, however they all define a direction that is mutually perpendicular to two other vectors. To find the direction of a moment, the first of these vectors is a position vector which points *from* the point of interest *to* a point on the force's line of action, and is designated $\mathbf{r}$. The second vector is the force, $\mathbf{F}$.

With the three-finger method, you align your right-hand index finger with vector $\mathbf{r}$ and your middle finger with the force $\mathbf{F}$; your thumb will point in the direction of the moment vector. Alternatively, you can align your thumb with $\mathbf{r}$ and your index finger with $\mathbf{F}$; your middle finger will point in the direction of the moment vector.



Figure 4.1.5 Three-finger right-hand rule for moments.

With the point-and-curl method, place your right hand flat and point your fingertips in the direction of $\mathbf{r}$. Rotate your hand until the force $\mathbf{F}$ is perpendicular to the back of your hand and can rotate your fingers. In this position, your thumb defines the direction of the moment vector and also the axis of rotation.

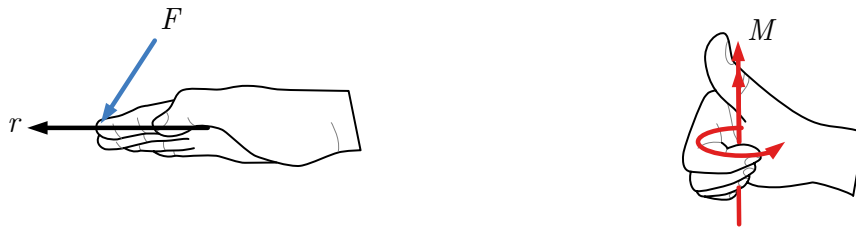


Figure 4.1.6 Point-and-curl right-hand rule for moments.

Any of these techniques may be used to find the direction of a moment. They all produce the same result so you don't need to learn them all, but make sure you have at least one method you can use accurately and consistently.

4.2 Scalar Addition of Moments

Since moments are vectors, they add according to the rules of vector addition. In three dimensions this usually involves finding the x , y , and z components of two or more moments and adding them to find the components of the resultant moment.

A simpler and more interactive approach using scalar components is available for moments in two dimensions.

4.2.1 Scalar Components

In [Subsection 3.3.2](#) we saw that vectors can be expressed as the product of a **scalar component** and a **unit vector**. For example, a 100 N force acting in the $-\mathbf{j}$ direction, i.e. straight down, can be represented by

$$\mathbf{F} = F_y \mathbf{j} = -100 \text{ N } \mathbf{j},$$

where $F_y = -100 \text{ N}$ is the scalar component and $\mathbf{j}$ is the unit vector. The absolute value of F_y determines the magnitude of the force, and unit vector along with the sign of F_y determines its direction. The negative sign here indicates that the direction of $\mathbf{F}$ is opposite to the direction of $\mathbf{j}$.

We have previously established two sign conventions for scalar components; we now add a third:

- For forces acting in the x direction, *right is positive*, corresponding to the direction of $\mathbf{i}$.
- For forces acting in the y direction, *up is positive*, corresponding to the direction of $\mathbf{j}$.
- For moments acting in the x - y plane, *counterclockwise moments are positive*, corresponding to the direction of $\mathbf{k}$ according to the right-hand rule.

Example 4.2.1 Sign Conventions.

For each scalar component, determine whether the moment is clockwise or counterclockwise, and apply the right-hand rule to determine the direction of the corresponding moment vector.

A $M_1 = 30 \text{ N}\cdot\text{m}$

B $M_2 = -400 \text{ kN}\cdot\text{m}$

C $M_3 = 25 \text{ N}\cdot\text{m} \curvearrowright$

D $M_4 = -100 \text{ ft}\cdot\text{lb} \curvearrowright$

Solution. By the right-hand rule, counterclockwise moments in the x - y plane are vectors that point in the $+\mathbf{k}$ direction. The corresponding scalar component has a positive sign. Clockwise moments point in the $-\mathbf{k}$ direction and have a negative sign.

A M_1 is positive, so it is CCW and the vector points in the $+\mathbf{k}$ direction.

B M_2 is negative, so it is CW and the vector points $-\mathbf{k}$ direction.

C The arrow on M_3 establishes the positive direction as CW. A positive scalar component means that the direction is CW and the vector points in the $-\mathbf{k}$ direction.

D The arrow on M_4 establishes the positive direction as CW, but the negative sign on the scalar component means that the actual moment is CCW. The vector points in the $+\mathbf{k}$ direction.

In three dimensions, moments, like forces, can be resolved into components in the x , y , and z directions.

$$\mathbf{M} = M_x \mathbf{i} + M_y \mathbf{j} + M_z \mathbf{k}.$$

This means that the three scalar components are required to fully specify a moment in three dimensions, whereas only one is required for moments in a plane.

Warning 4.2.2

Be careful not to mix up magnitudes with scalar components.

- Both are scalar values with units.
- Magnitudes are never negative. Scalar components may be positive or negative.

- Scalar components always have an associated sign convention. It may be implied or specifically indicated. If the sign convention is unstated, assume that counterclockwise moments in the x - y plane are positive.
- There is no special symbol or notation to indicate whether a quantity represents a vector magnitude or a scalar moment, so pay attention to context.

4.2.2 Addition of Moments

Scalar components are most useful for adding and finding the resultant of several moments. When moments are expressed in terms of scalar components, vector addition reduces to algebraic addition of the scalar components.

To add several moments in a plane using scalar components, simply append a positive sign to the magnitude of counterclockwise moments and a negative sign to clockwise moments to create scalar components, then add the scalar components normally.

The resulting algebraic sum of the scalar components will be either positive, negative, or zero, and this sign indicates the direction of the resultant moment. If the result is positive, the resultant acts in the direction of the positive unit vector, and in the opposite direction if it is negative.

Example 4.2.3 Scalar addition.

Find the resultant of the following three moments:

$\mathbf{M}_1 = 25 \text{ kN}\cdot\text{m} \curvearrowright$, $\mathbf{M}_2 = 40 \text{ kN}\cdot\text{m} \curvearrowleft$, and $\mathbf{M}_3 = 30 \text{ kN}\cdot\text{m} \curvearrowright$

Solution. Manually attaching the positive or negative signs according to the right-hand rule turns the magnitudes into scalar moments:

$$M_1 = -25 \text{ kN}\cdot\text{m}$$

$$M_2 = +40 \text{ kN}\cdot\text{m}$$

$$M_3 = -30 \text{ kN}\cdot\text{m}$$

Adding these moments algebraically gives the resultant scalar moment.

$$\begin{aligned} M &= M_1 + M_2 + M_3 \\ &= (-25 \text{ kN}\cdot\text{m}) + (40 \text{ kN}\cdot\text{m}) + (-30 \text{ kN}\cdot\text{m}) \\ &= -15 \text{ kN}\cdot\text{m} \end{aligned}$$

The negative sign indicates that the direction of the resultant moment is clockwise.

Interpreting the scalar moment as a vector gives:

$$\mathbf{M} = 15 \text{ kN}\cdot\text{m} \curvearrowleft$$

The corresponding magnitude of $\mathbf{M}$ is

$$|\mathbf{M}| = 15 \text{ kN}\cdot\text{m}$$

4.3 Varignon's Theorem

Varignon's Theorem is a method to calculate moments developed in 1687 by French mathematician Pierre Varignon (1654 – 1722). It states that

The sum of the moments of several concurrent forces about a point is equal to the moment of the resultant of those forces, or alternately, the moment of a force about a point equals the sum of the moments of its components.

This means you can find the moment of a force by first breaking it into components, evaluating the moments of the individual components, and finally summing them to find the net moment about the point.

This may sound like more work than just finding the moment of the original force, but in practice, it is often easier.

When applying Varignon's Theorem, you are free to select components in any direction, but the two most convenient choices are discussed below.

4.3.1 Parallel and Perpendicular Components

The first common way to apply Varignon's Theorem is to break the force creating the moment into two components, one parallel with the line segment connecting the moment center with the point of application of the force, and the other perpendicular to it.

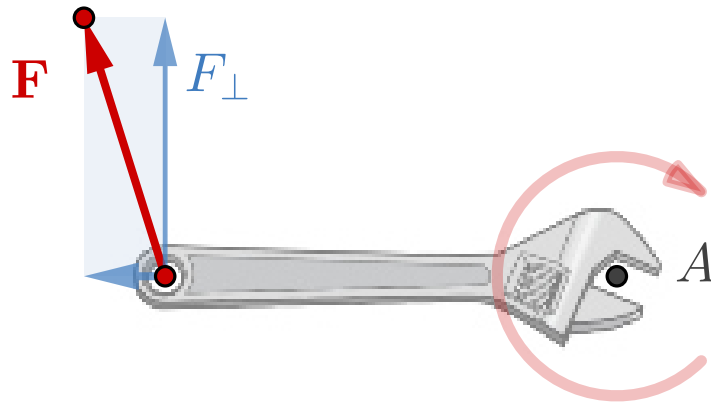
Consider the wrench in the interactive diagram below. Assume that the length of the handle is d , the magnitude of the force is F , and the angle between the force and the handle is θ . The components parallel to the wrench handle and perpendicular to it are

$$F_{\parallel} = F \cos \theta \quad F_{\perp} = F \sin \theta.$$

By Varignon's theorem and the definition of the moment, (4.1.1), the sum of the moments of the components simplifies to:

$$M = F_{\perp} d, \tag{4.3.1}$$

because the contribution of the parallel component is zero since its line of action passes through the moment center A .



[Live Link](#)

Figure 4.3.1 Varignon's Theorem: Perpendicular Component

This result agrees with our intuitive understanding of how a wrench works; the greatest torque is developed when the force is applied at a right angle to the handle.

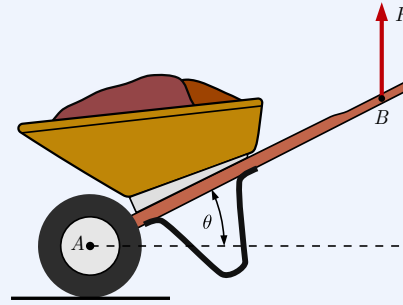
Equations (4.1.1) and (4.3.1) not only produce the same result, they are completely identical. If the length of the handle is d and the angle between the force $\mathbf{F}$ and the handle is θ , then $d_{\perp} = d \sin \theta$, and $F_{\perp} = F \sin \theta$. Using either equation to calculate the moment gives

$$M = \begin{cases} F d_{\perp} & = F (d \sin \theta) \\ F_{\perp} d & = (F \sin \theta) d \end{cases} = Fd \sin \theta. \quad (4.3.2)$$

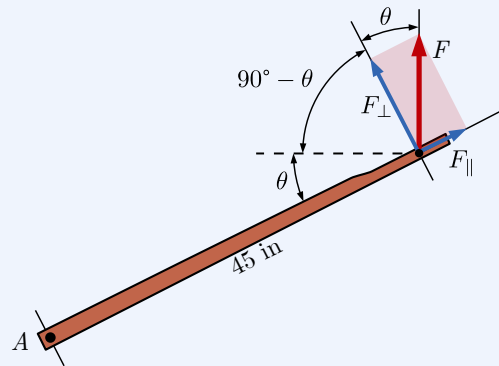
Example 4.3.2 Varignon's Theorem: Parallel and Perpendicular Components.

A gardener is moving soil with a wheelbarrow. The length of the handle from A to B is 45 in, and in the position shown, θ is 25° .

Find the moment created about axle A when she applies a vertical 30 lb force at B by resolving force F into components parallel and perpendicular to the handle.



Solution.



When the handle makes an angle of θ with the horizontal axis, the perpendicular component of F makes the same angle with the vertical, so resolving F into perpendicular and parallel components gives:

$$F_{\perp} = F \cos \theta = 27.19 \text{ lb} \qquad F_{\parallel} = F \sin \theta = 12.68 \text{ lb}$$

Summing the moments of the components we find

$$M_A = F_{\perp}(45 \text{ in}) + F_{\parallel}(0) = 1224 \text{ in}\cdot\text{lb}$$

Note that the parallel component does not contribute to the moment.

4.3.2 Horizontal and Vertical Components

Varignon's theorem is particularly convenient in situations where horizontal and vertical dimensions are provided, as is often the case. If you decompose the forces into horizontal and vertical components, you can find the moments of the components without difficulty. The resultant moment is the sum of the moments of the components.

Varignon's Equation is (4.3.3) and Figure 4.3.3 is the corresponding diagram

for the moment of force F about point A . Notice that the horizontal component is multiplied by the vertical distance, and the vertical component is multiplied by the horizontal distance since these are the perpendiculars, and that the signs on the terms can be either positive or negative, depending on the direction of the component moments.

$$M = \pm(F_x d_y) \pm (F_y d_x) \quad (4.3.3)$$

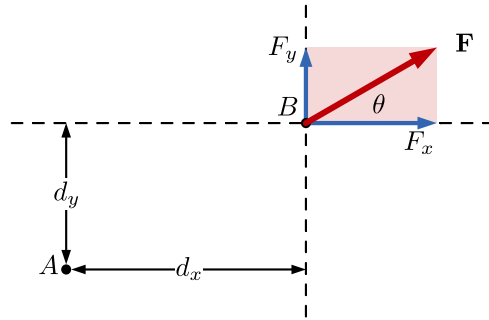
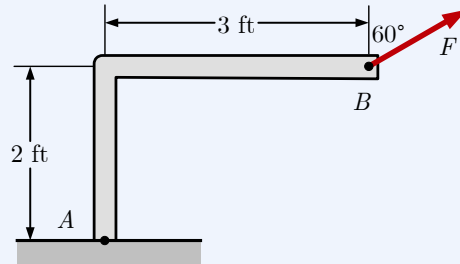


Figure 4.3.3 Varignon's Theorem Horizontal and Vertical Components

When applying Varignon's Theorem (4.3.3), take care to assign the correct sign to the individual moment terms; counterclockwise moments are positive, and clockwise negative are negative according to the right-hand rule, regardless of whether F_x acts left or right, or whether F_y acts up or down.

Example 4.3.4 Varignon's Theorem: Horizontal and Vertical Components.

A 750 lb force is applied to the frame as shown. Determine the moment this force makes about point A by resolving the force into horizontal and vertical components and applying Varignon's Theorem.



Solution. Force $\mathbf{F}$ acts 60° from the vertical with a 750 lb magnitude, so its horizontal and vertical components are

$$F_x = F \sin 60^\circ = 649.5 \text{ lb}$$

$$F_y = F \cos 60^\circ = 375.0 \text{ lb}$$

For component F_x , the perpendicular distance from point A is 2 ft so the moment of this component is

$$M_1 = 2F_x = 1299 \text{ ft}\cdot\text{lb Clockwise.}$$

For component F_y , the perpendicular distance from point A is 3 ft so the moment of this component is

$$M_2 = 3F_y = 1125 \text{ ft}\cdot\text{lb counterclockwise.}$$

Assigning a negative sign to M_1 and a positive sign to M_2 to account for their directions and summing, gives the moment of $\mathbf{F}$ about A .

$$\begin{aligned}M_A &= -M_1 + M_2 \\&= -1299 + 1125 \\&= -174 \text{ ft}\cdot\text{lb}\end{aligned}$$

The negative sign indicates that the resultant moment is clockwise, with a magnitude of 174 ft·lb.

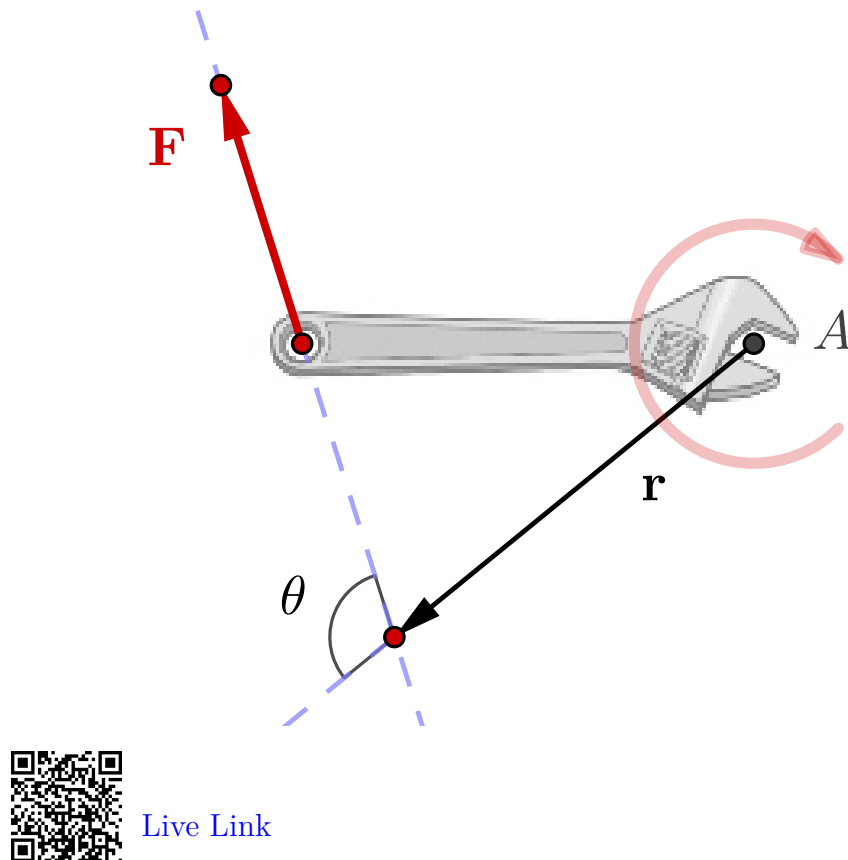
$$\mathbf{M}_A = 174 \text{ ft}\cdot\text{lb} \text{ Clockwise.}$$

4.3.3 Moment Cross Products

The most robust and general method to find a moment of a force is to use the vector cross product

$$\mathbf{M} = \mathbf{r} \times \mathbf{F}, \quad (4.3.4)$$

where $\mathbf{F}$ is the force creating the moment, and $\mathbf{r}$ is a position vector from the moment center to any point on the line of action of the force, as shown in [Figure 4.3.5](#). The cross product is a vector multiplication operation, and the result is a vector perpendicular to the two vectors being multiplied.



[Live Link](#)

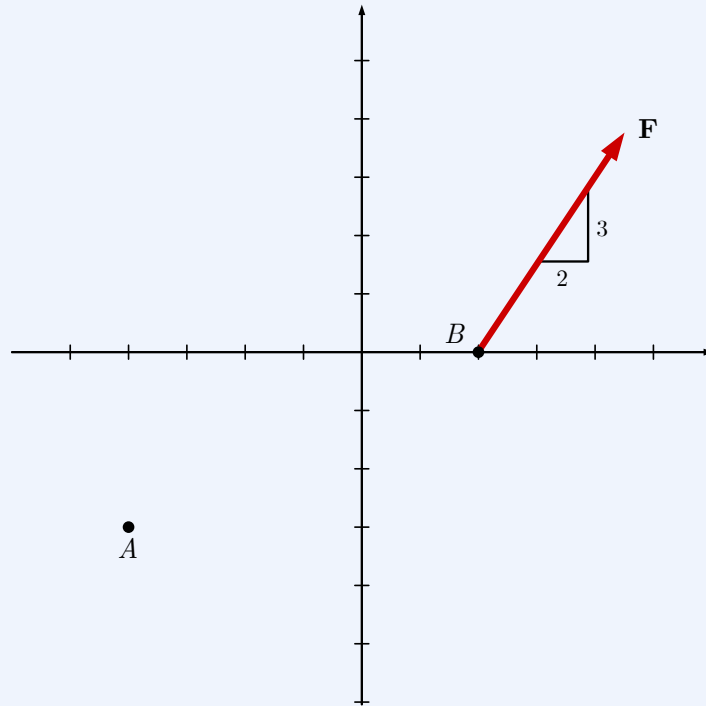
Figure 4.3.5 Moment cross product. $\mathbf{M} = \mathbf{r} \times \mathbf{F}$

The cross product method is most often used to find moments in three dimensions, as will be discussed in [Section 4.4](#), but it can also be applied in two-dimensional problems.

The next example demonstrates the technique and compares it to the other methods we have already discussed. You will see that the cross product method is mathematically identical to applying Varignon's theorem.

Example 4.3.6 2D Moments - Four Ways.

This problem demonstrates four different ways you can find the moment of a force about a point. The first two methods use vector algebra; the second two take a scalar approach that uses geometry and right-triangle trigonometry. All four methods are mathematically identical.



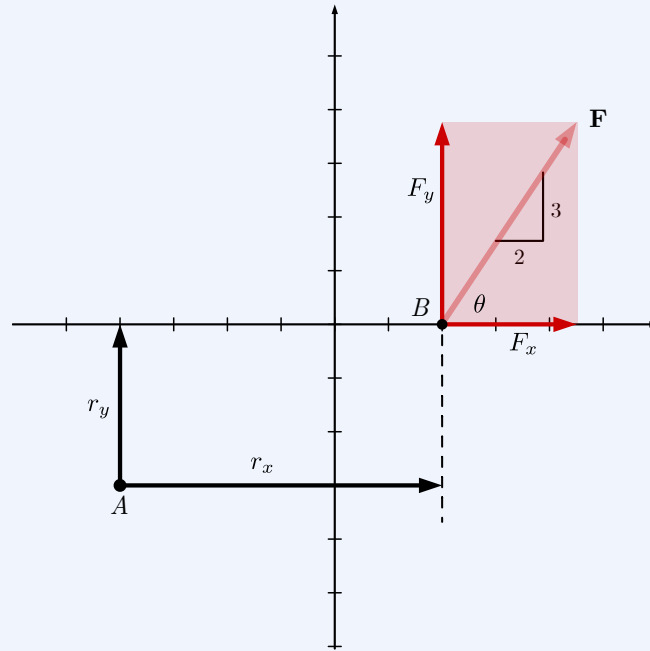
Force $\mathbf{F}$ has a magnitude of 500 lbf and acts on point B in the direction shown. Find the moment caused by force $\mathbf{F}$ around point $A = (-4, -3)$ ft using different methods and verify that they give the same result.

- (a) Find the moment of $\mathbf{F}$ about point A using Varignon's Theorem,

$$\mathbf{M}_A = (\mathbf{r}_x \times \mathbf{F}_y) + (\mathbf{r}_y \times \mathbf{F}_x).$$

Answer. $\mathbf{M}_A = 1664.10 \text{ ft}\cdot\text{lbf} (+\mathbf{k})$

Solution. [Varignon's Theorem](#) states that the moment of a force is the sum of the moments of its components. In this example, we will determine the vertical and horizontal components of $\mathbf{r}$ and $\mathbf{F}$, then add the cross products of the two perpendicular pairs.



The 3:2 slope of $\mathbf{F}$ can be expressed as an angle.

$$\theta = \tan^{-1} \frac{3}{2} = 56.3^\circ$$

Find the components of $\mathbf{r}$ and $\mathbf{F}$.

$$\begin{aligned}\mathbf{F} &= 500 \text{ lbf} \langle \cos 56.3^\circ, \sin 56.3^\circ \rangle \\ &= \langle 277.35, 416.025 \rangle \text{ lbf} \\ \mathbf{r} &= \langle 6, 3 \rangle \text{ ft}\end{aligned}$$

Finally, following Varignon's Theorem, add the cross products of the perpendicular component pairs.

$$\begin{aligned}\mathbf{M}_A &= (\mathbf{r}_x \times \mathbf{F}_y) + (\mathbf{r}_y \times \mathbf{F}_x) \\ &= 6 \text{ ft} \cdot 416.025 \text{ lbf} (\mathbf{k}) + 3 \text{ ft} \cdot 277.35 \text{ lbf} (-\mathbf{k}) \\ &= 1664.1 \text{ ft} \cdot \text{lbf} (+\mathbf{k})\end{aligned}$$

Notes:

- When finding the moment of two-dimensional vectors in component form, this is often the preferred method, as it is quick and most find the process intuitive.
- The first cross product, $\mathbf{r}_x \times \mathbf{F}_y$, has a *positive value* because $\mathbf{i} \times \mathbf{j} = +\mathbf{k}$, *not* because you are simply multiplying two positive components.

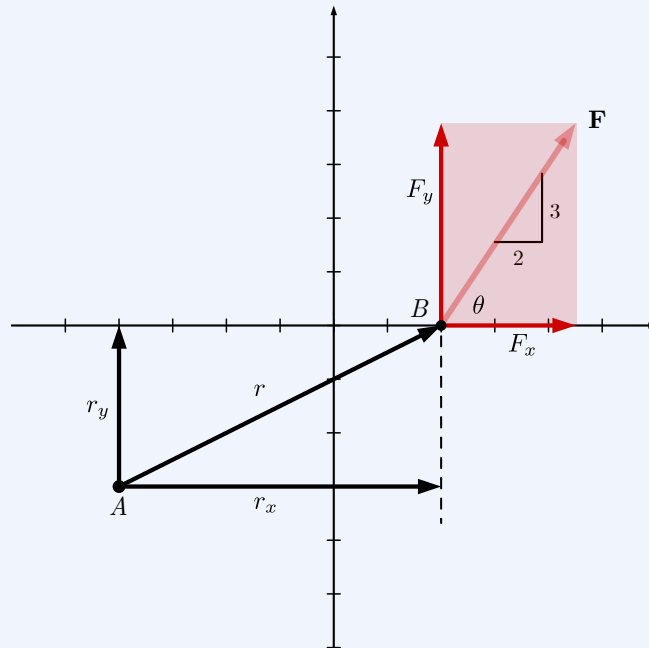
- The second cross product, $\mathbf{r}_y \times \mathbf{F}_x$, results in a *negative value* because $\mathbf{j} \times \mathbf{i} = -\mathbf{k}$.
- All moments have units of force times distance, in this case [ft · lbf].
- The overall sign of $\mathbf{M}_A$ determines the final direction. A positive value corresponds to a counterclockwise moment – right thumb out of the page – and a negative value indicates a clockwise moment. See Figure 4.1.6 for the hand diagram.

(b) Find the moment of $\mathbf{F}$ about point A using a vector cross product,

$$\mathbf{M}_A = \mathbf{r} \times \mathbf{F}.$$

Answer. $\mathbf{M}_A = 1664.10 \text{ ft} \cdot \text{lbf} (+\mathbf{k})$

Solution. We can also solve for the moment $\mathbf{M}_A$ using the vector determinant method of Subsection 2.8.3. We can use the values of θ , $\mathbf{r}$, and $\mathbf{F}$ computed in part (a) above.



Using the the vector determinant, we find:

$$\begin{aligned} \mathbf{M}_A &= \mathbf{r} \times \mathbf{F} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & 0 \\ F_x & F_y & 0 \end{vmatrix} \end{aligned}$$

$$\begin{aligned}
&= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 6 & 3 & 0 \\ 277.35 & 416.025 & 0 \end{vmatrix} \\
&= 6 \text{ ft} (416.025 \text{ lbf}) (\mathbf{k}) + 3 \text{ ft} (277.35 \text{ lbf}) (-\mathbf{k}) \\
&= 1664.1 \text{ ft}\cdot\text{lbf} (+\mathbf{k})
\end{aligned}$$

Notes:

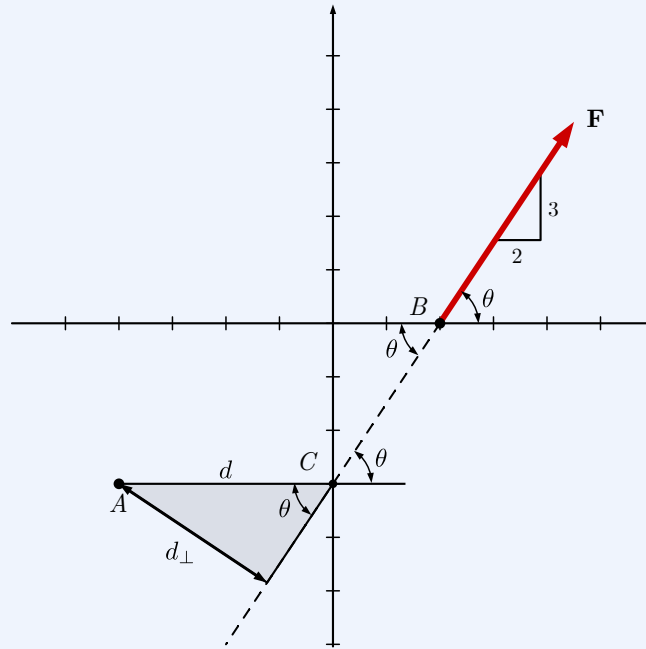
- Determinants are a robust way to compute two-dimensional cross products, but take a bit more effort than Varignon's Theorem. The math is *exactly the same* either way, which means that Varignon's Theorem is just a shortcut to working through a two-dimensional vector determinant.
- The signs on the cross-product terms $\mathbf{r}_x \times \mathbf{F}_y$ and $\mathbf{r}_y \times \mathbf{F}_x$ still come from the right-hand rule, and conveniently the process of multiplying diagonals in the determinant takes care of the signs.
- Recognize that the reason we multiply diagonals in a determinant is that we only want to multiply the *perpendicular* components.

- (c) Find the moment of $\mathbf{F}$ about point A by finding the perpendicular distance $d_\perp$,

$$M_A = F d_\perp.$$

Answer. $\mathbf{M}_A = 1664.10 \text{ ft}\cdot\text{lbf} (+\mathbf{k})$

Solution. This solution requires you to find the perpendicular distance $d_\perp$ between the point A and line-of-action of $\mathbf{F}$. One way to find this distance is shown below.



- (a) Draw a moderately large and accurate diagram. Too much confusion has been created by small, inaccurately-drawn diagrams. Include a right triangle whose hypotenuse connects the moment center A with a convenient point on the line of action of $\mathbf{F}$. We have chosen point $C = (0, -3)$, although point B would work as well.

Can you prove that C is on the line of action?

- (b) Since the angle θ between the line of action of the force and any horizontal line is the same everywhere, θ appears three more times on this diagram, and gives us an angle in the right triangle.
- (c) Use right triangle trigonometry to find $d_{\perp}$.

$$\begin{aligned}\sin \theta &= \frac{d_{\perp}}{AC} \\ d_{\perp} &= AC (\sin \theta) \\ &= 4 (\sin 56.3^{\circ}) \\ &= 3.328 \text{ ft}\end{aligned}$$

- (d) Finally, compute the moment about A .

$$\begin{aligned}M_A &= F d_{\perp} \\ &= 3.328 \text{ ft} (500 \text{ lbf}) \\ &= 1664.10 \text{ ft}\cdot\text{lbf} \\ \mathbf{M}_A &= 1664.10 \text{ ft}\cdot\text{lbf} (+\mathbf{k})\end{aligned}$$

The $(+\mathbf{k})$ direction of $\mathbf{M}_A$ comes from the observation of the right-hand rule, as scalar moment computations are not directional.

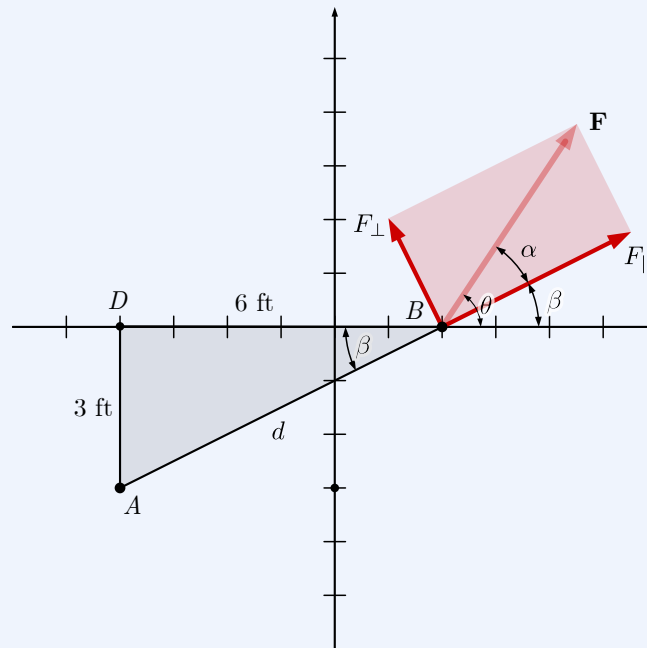
- (d) Find the moment of $\mathbf{F}$ about point A by finding the perpendicular component of $\mathbf{F}$,

$$M_A = F_{\perp} d.$$

Answer. $\mathbf{M}_A = 1664.10 \text{ ft}\cdot\text{lbf} (+\mathbf{k})$

Solution. This solution requires you to find the rectangular component of force $\mathbf{F}$ perpendicular to segment d connecting the moment center A with a point on the line of action of $\mathbf{F}$.

One approach to finding $F_{\perp}$ is shown below.



- (a) Draw a large and accurate diagram to assist in finding the distances and angles in this problem.

Draw a line through point A and the point where $\mathbf{F}$ acts, or to any other convenient point on its line of action. Then draw a rectangle aligned with this line, indicating the parallel and perpendicular components of $\mathbf{F}$. This rectangle consists of two congruent right triangles, having the force as their common hypotenuse. You will need to find an angle inside the triangle to find the components.

- (b) Use triangle ABD to compute angle β and the length of the hypotenuse d .

$$\beta = \tan^{-1} \left(\frac{3}{6} \right) = 26.565^\circ$$

$$d = \sqrt{6^2 + 3^2} = 6.708 \text{ ft}$$

- (c) Recalling that we have previously found $\theta = 56.31^\circ$, find α .

$$\alpha = \theta - \beta = 29.74^\circ$$

- (d) Find $\mathbf{F}_\perp$ using the force triangle.

$$\begin{aligned} F_\perp &= F \sin \alpha \\ &= 500 \text{ lbf} \sin 29.74^\circ \\ &= 248.07 \text{ lbf} \end{aligned}$$

- (e) Finally, compute the magnitude M_A and the vector $\mathbf{M}_A$.

$$\begin{aligned} M_A &= F_\perp d \\ &= 248.07 \text{ lbf} \cdot 6.708 \text{ ft} \\ \mathbf{M}_A &= 1664.10 \text{ ft} \cdot \text{lbf} (+\mathbf{k}) \end{aligned}$$

The counterclockwise direction $(+\mathbf{k})$ comes from the right-hand rule, since scalar moment computations are not directional.

We have discussed several different approaches for finding the moment of a force about a point in the preceding sections. While all of these methods produce the same result, you will frequently find that one method is computationally simpler to apply than another. The simplest method generally depends on what is known and unknown in a given situation, so we recommend that you become familiar with all these methods.

The interactive diagram below will help you visualize the different approaches for finding moments that were covered.

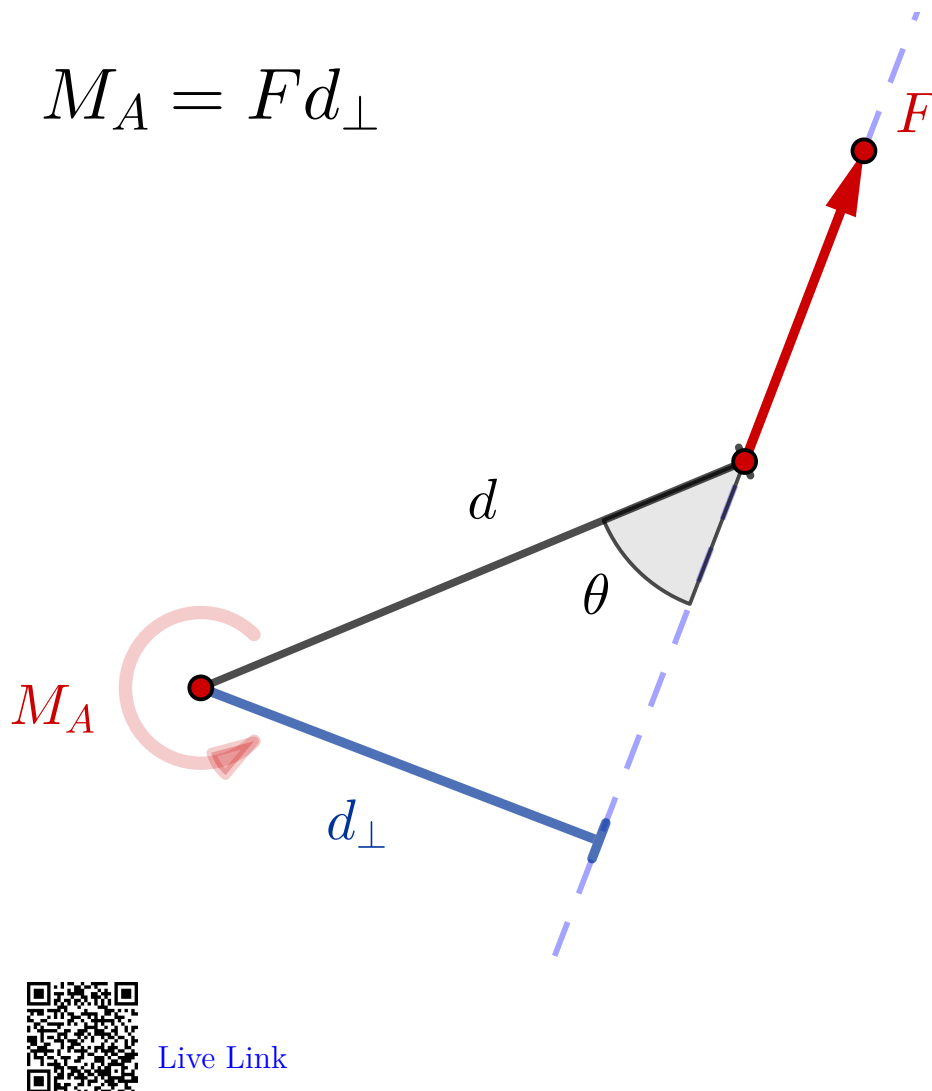


Figure 4.3.7 Four equivalent approaches to finding a moment about a point.

4.4 3D Moments

Key Questions

- Where does the moment arm vector $\mathbf{r}$ start and end?
- Why does Varignon's Theorem give you the same answer as a determinant?
- How can you combine a dot product and a cross product to find the moment about a line?
- Why does a mixed-triple determinant give you a scalar while a cross-

product determinant gives you a vector?

The circular arrows we used to represent vectors in two dimensions are unclear in three dimensions, so instead, moments are represented as arrows with x , y , and z components, like force and position vectors. You will sometimes see moments indicated with double arrowheads to differentiate them from force vectors.

In three dimensions, it is usually not convenient to use geometry to find the moment arm for equation (4.1.1), so instead we will use the vector cross product method, which is easier to apply but less intuitive.

4.4.1 3D Cross Products

The mathematics of cross products was discussed in Section 2.8, and equation (2.8.1) provides one method to calculate a moment cross products

$$\mathbf{M} = |\mathbf{r}||\mathbf{F}| \sin \theta \hat{\mathbf{u}} \quad (4.4.1)$$

. Here, θ is the angle between the two vectors as shown in Figure 4.3.5 above, and $\hat{\mathbf{u}}$ is the unit vector perpendicular to both $\mathbf{r}$ and $\mathbf{F}$ with the direction coming from the right-hand rule. This equation is useful if you know or can find the magnitudes of $\mathbf{r}$ and $\mathbf{F}$ and the angle θ between them. This equation is the vector equivalent of (4.3.2).

Alternatively, if you know or can find the components of the position $\mathbf{r}$ and force $\mathbf{F}$ vectors, it's typically easiest to evaluate the moment cross product using the determinant form discussed in Subsection 2.8.3.

$$\begin{aligned} \mathbf{M} &= \mathbf{r} \times \mathbf{F} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix} \\ &= (r_y F_z - r_z F_y) \mathbf{i} - (r_x F_z - r_z F_x) \mathbf{j} + (r_x F_y - r_y F_x) \mathbf{k} \end{aligned} \quad (4.4.2)$$

Here, r_x , r_y , and r_z are components of the vector describing the distance from the point of interest to the force. F_x , F_y , and F_z are components of the force. The resulting moment has three components.

$$\begin{aligned} M_x &= (r_y F_z - r_z F_y) \\ M_y &= -(r_x F_z - r_z F_x) \\ M_z &= (r_x F_y - r_y F_x) \end{aligned}$$

These represent the component moments acting around each of the three coordinate axes. Note the negative sign in front of M_y , which ensures that the cross products maintain the correct signs. The magnitude of the resultant moment can be calculated using the three-dimensional Pythagorean Theorem.

$$M = |\mathbf{M}| = \sqrt{M_x^2 + M_y^2 + M_z^2} \quad (4.4.3)$$

It is important to avoid three common mistakes when setting up the cross product.

- The order must always be $\mathbf{r} \times \mathbf{F}$, never $\mathbf{F} \times \mathbf{r}$. The moment arm $\mathbf{r}$ appears in the middle line of the determinant and the force $\mathbf{F}$ on the bottom line.
- The moment arm $\mathbf{r}$ must always be measured from the moment center to the line of action of the force. Never from the force to the point.
- The signs of the components of $\mathbf{r}$ and $\mathbf{F}$ must follow those of a right-hand coordinate system.

In two dimensions, r_z and F_z are zero, so (4.4.2) reduces to

$$\begin{aligned} \mathbf{M} &= \mathbf{r} \times \mathbf{F} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & 0 \\ F_x & F_y & 0 \end{vmatrix} \\ &= (r_x F_y - r_y F_x) \mathbf{k} \end{aligned} \quad (4.4.4)$$

. This is just the vector equivalent of [Varignon's Theorem](#) in two dimensions, with the correct signs automatically determined from the signs on the scalar components of $\mathbf{F}$ and $\mathbf{r}$.

4.4.2 Moment about a Point

The next two interactives should help you visualize moments in three dimensions.

The first shows the force vector, position vector, and the resulting moment all placed at the origin for simplicity. The moment is perpendicular to the plane containing $\mathbf{F}$ and $\mathbf{r}$ and has a magnitude equal to the 'area' of the parallelogram with $\mathbf{F}$ and $\mathbf{r}$ for sides.

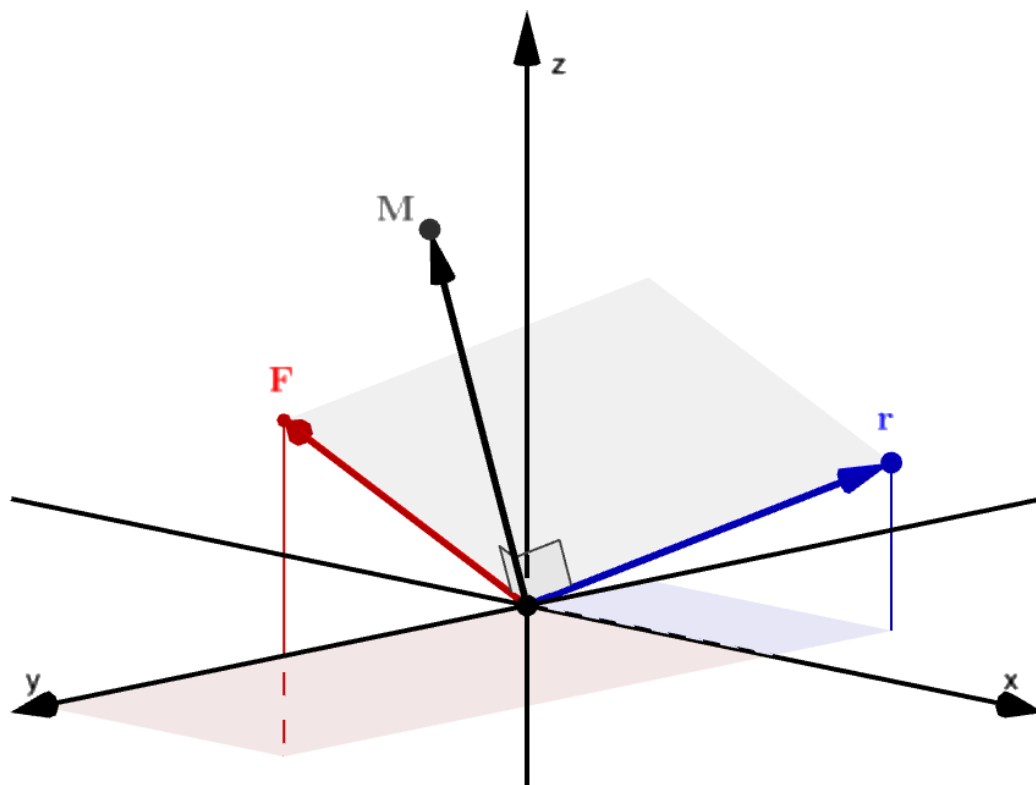
[Live Link](#)

Figure 4.4.1 Moment about the origin.

The second interactive shows a more realistic situation. The moment center is at an arbitrary point A , and the line of action of force $\mathbf{F}$ passes through arbitrary points P_1 and P_2 . The position vector $\mathbf{r}$ is the vector from A to a point on the line of action, and the force $\mathbf{F}$ can be slid anywhere along that line.

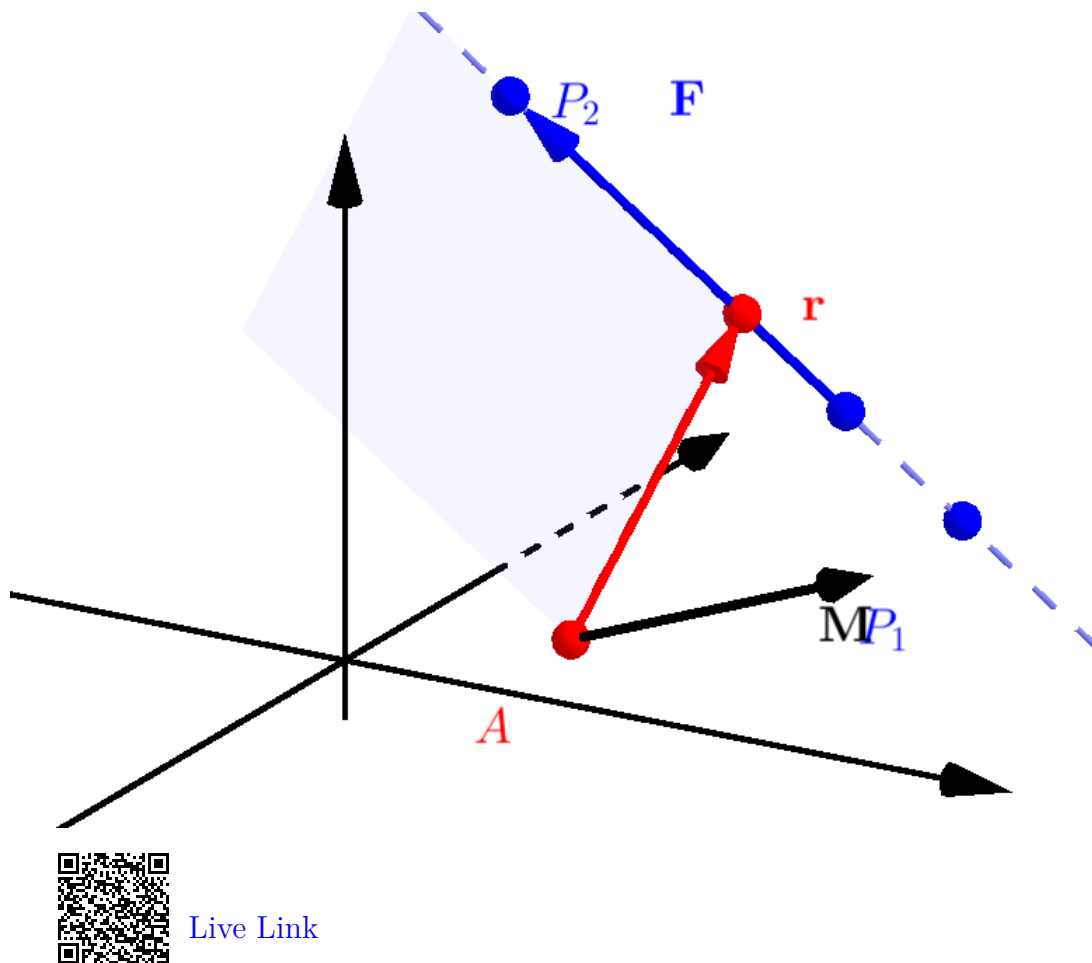
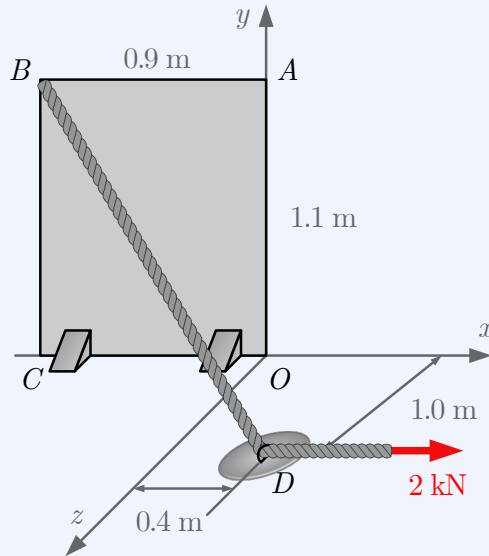


Figure 4.4.2 Moment about an arbitrary point.

Example 4.4.3 3D Moment about a Point.

A thin plate $OABC$ sits in the xy plane. Cable BD pulls with a tension of 2 kN through a frictionless ring at point D . Find the moment caused by the tension force around point O .



Solution. Start the problem by using the position information and tension magnitude to find the force vector $\mathbf{F}_{BD}$. This will be done in three steps:

1. *Find the position vector $\mathbf{BD}$:* Find position vectors by either subtracting the start-point coordinates from the end-point coordinates or focusing on the changes in the position components from B to D .

$$\begin{aligned}\mathbf{BD} &= \mathbf{D} - \mathbf{B} \\ &= (0.4, 0, 1.0) \text{ m} - (-0.9, 1.1, 0) \text{ m} \\ &= \langle 1.3, -1.1, 1 \rangle \text{ m}\end{aligned}$$

2. *Find the unit vector of $\mathbf{BD}$:* Compute a unit vector by dividing $\mathbf{BD}$ by the total length of BD .

$$\begin{aligned}BD &= |\mathbf{BD}| = \sqrt{1.3^2 + (-1.1)^2 + 1.0^2} \\ &= 1.975 \text{ m} \\ \widehat{\mathbf{BD}} &= \frac{\mathbf{BD}}{BD} \\ &= \frac{\langle 1.3, -1.1, 1 \rangle \text{ m}}{1.975 \text{ m}} \\ &= \langle 0.658, -0.557, 0.506 \rangle\end{aligned}$$

Note that $\widehat{\mathbf{BD}}$ is unitless and is the pure direction of $\mathbf{BD}$.

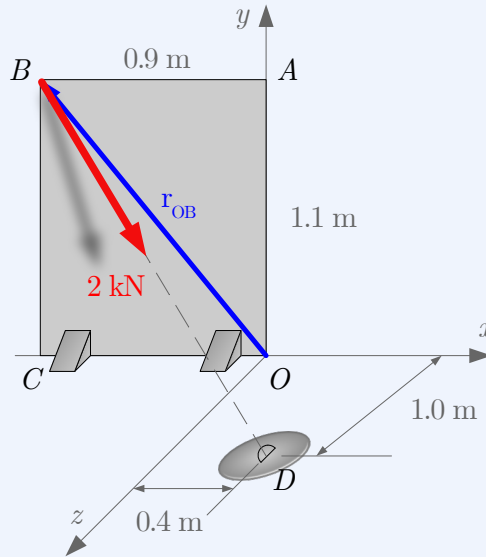
3. *Multiply the unit vector by force magnitude:* Now multiply $\widehat{\mathbf{BD}}$ by the 2 kN force magnitude to find the force components.

$$\mathbf{F}_{BD} = F_{BD}(\widehat{\mathbf{BD}})$$

$$\begin{aligned}
 &= 2 \text{ kN} \langle 0.658, -0.557, 0.506 \rangle \\
 &= \langle 1.317, -1.114, 1.013 \rangle \text{ kN}
 \end{aligned}$$

Next, find the moment arm from point O to the line of action of the force. There are two obvious options for moment arms, either $\mathbf{r}_{OB}$ or $\mathbf{r}_{OD}$. To demonstrate how both moment arms give the same answer, solutions for both moment arms will be shown.

Option 1: Moment using $\mathbf{r}_{OB}$



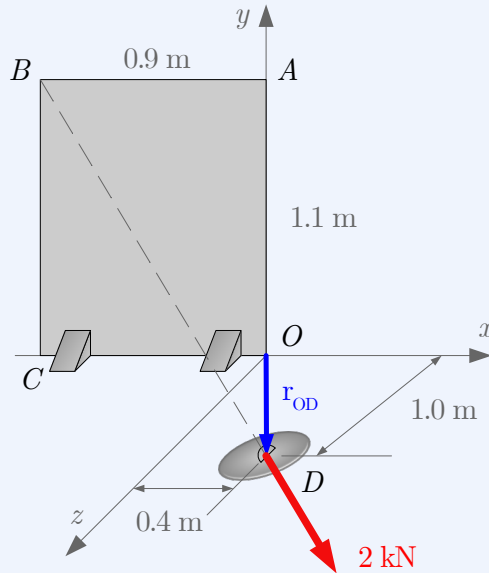
- Moment arm $\mathbf{r}_{OB}$ starts at the point we are taking the moment around, O , and ends at the point B .

$$\begin{aligned}
 \mathbf{OB} &= B - O \\
 &= (-0.9, 1.1, 0) \text{ m} - (0, 0, 0) \text{ m} \\
 &= \langle -0.9, 1.1, 0 \rangle \text{ m}
 \end{aligned}$$

- Cross $\mathbf{r}_{OB}$ with $\mathbf{F}_{BD}$ to find the moment of $\mathbf{F}_{BD}$ about point O .

$$\begin{aligned}
 \mathbf{M}_O &= \mathbf{r}_{OB} \times \mathbf{F}_{BD} \\
 &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.9 & 1.1 & 0 \\ 1.317 & -1.114 & 1.013 \end{vmatrix} \\
 &= \langle 1.114, 0.911, -0.446 \rangle \text{ kN}
 \end{aligned}$$

Option 2: Moment using $\mathbf{r}_{OD}$:



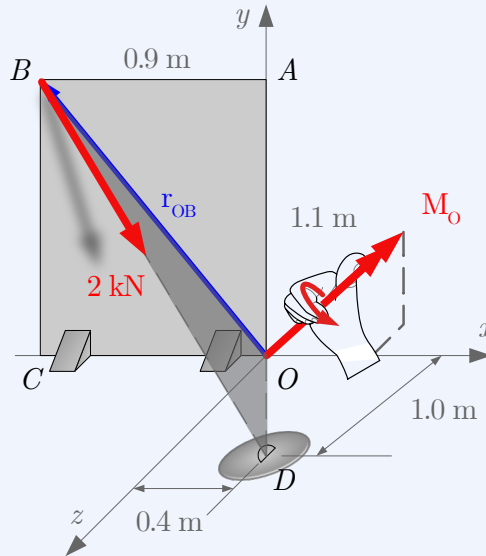
- Moment arm $\mathbf{r}_{OD}$ starts at the point we are taking the moment around, O , and ends at the point D .

$$\begin{aligned}\mathbf{OD} &= D - O \\ &= (0.4, 0, 1.0) \text{ m} - (0, 0, 0) \text{ m} \\ &= \langle 0.4, 0, 1.0 \rangle \text{ m}\end{aligned}$$

- Cross $\mathbf{r}_{OD}$ with $\mathbf{F}_{BD}$ to find the moment of $\mathbf{F}_{BD}$ about point O .

$$\begin{aligned}\mathbf{M}_O &= \mathbf{r}_{OD} \times \mathbf{F}_{BD} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.4 & 0 & 1.0 \\ 1.317 & -1.114 & 1.013 \end{vmatrix} \\ &= \langle 1.114, 0.911, -0.446 \rangle \text{ kN}\end{aligned}$$

It is worth your effort to compute moments both ways for this example, or another problem, to prove to yourself that the answers work out exactly the same with different moment arms. You can actually select a moment arm from O to *any* point on line $\mathbf{BD}$ and get the same result, but B and D are points on that line that we already know.

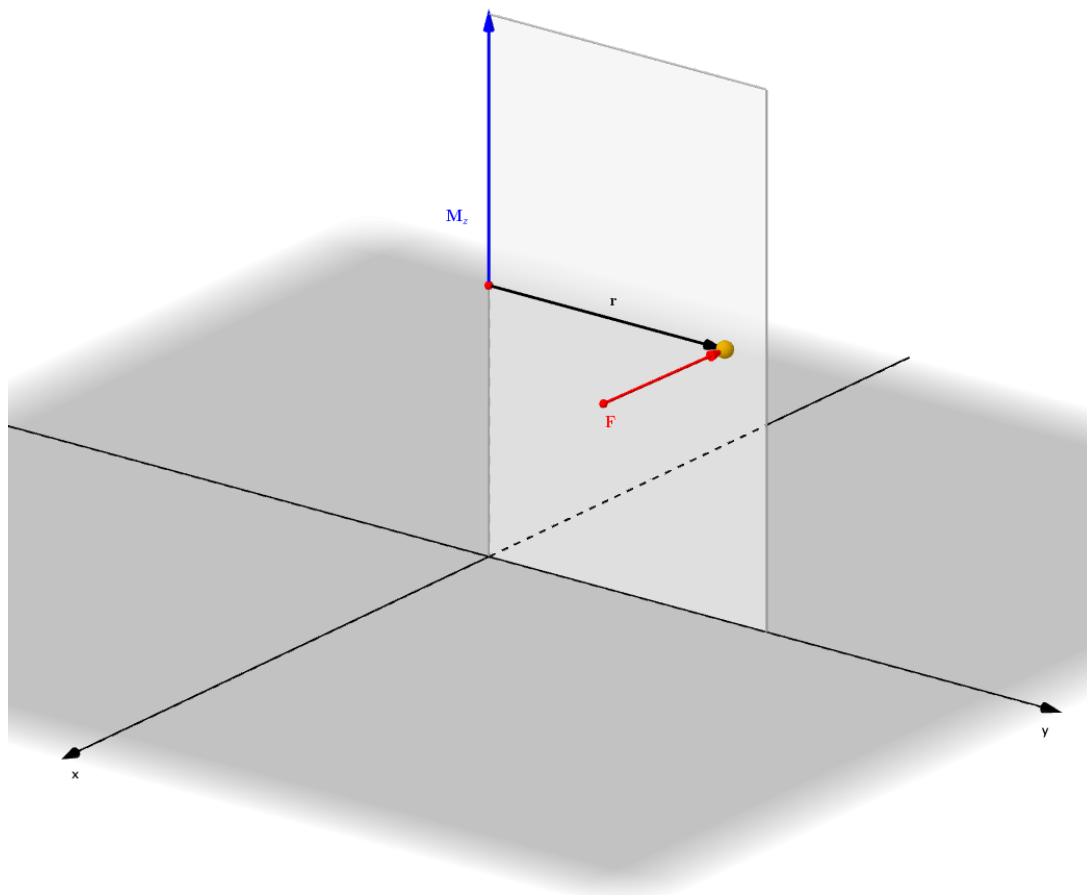


Drawing $\mathbf{M}_O$, demonstrates that a moment vector direction is both 1) the axis of rotation caused by $\mathbf{T}_{BD}$ around point O , with the moment aligning to your thumb and the moment rotating around your fingers from the right-hand rule and 2) that $\mathbf{M}_O$ is perpendicular to the plane formed by $\mathbf{T}_{BD}$ and $\mathbf{T}_{BD}$. Recall that *all* cross products result in vectors perpendicular to the two crossed vectors.

4.4.3 Moment about a Line

In three dimensions, the moment of a force about a point can be resolved into components about the x , y and z axes. The moment produces a rotational tendency about all three axes simultaneously, but only a portion of the total moment acts about any particular axis.

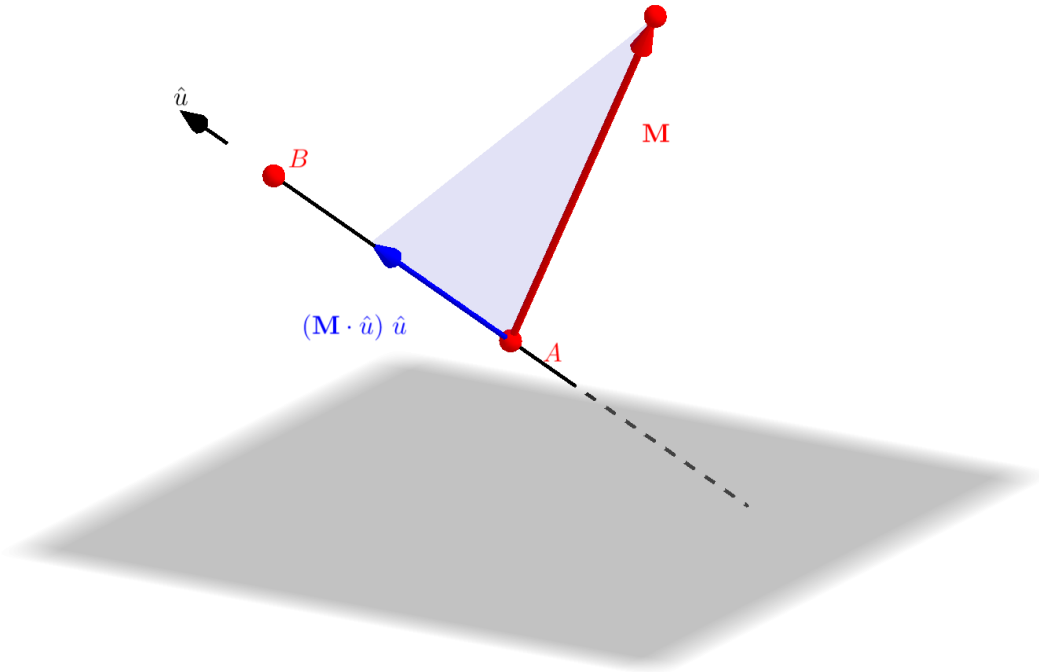
We are often interested in finding the effect of a moment about a specific line or axis. For example, consider the moment created by a push on a door handle. Unless you push with a force exactly perpendicular to the hinge, only a portion of the total moment you produce will act around the hinge axis and be effective to open the door. The moment we are looking for is the vector projection of the moment onto the axis of interest. Vector projections were first discussed in [Subsection 2.7.4](#).



[Live Link](#)

Figure 4.4.4 Moment on a hinge.

The axis of interest does not need to be a coordinate axis. This interactive shows the projection of moment $\mathbf{M}$ on a line passing through points A and B .



[Live Link](#)

Figure 4.4.5 Moment of a force about a line

To find the moment of a force about a line, three vectors are required:

- $\hat{\mathbf{u}}$, a unit vector pointing in the direction of the line or axis of interest.
- $\mathbf{r}$, a position vector from any point on the line of interest to any point on the line of action of the force.
- $\mathbf{F}$, the force vector. If you have multiple concurrent forces, you can treat them individually or add them together first and find the moment of the resultant.

Using these three vectors, we can compute the magnitude of the moment about a line in two related, but unique ways.

Two-Step Method: Separates the cross and dot products into two distinct steps.

- First, compute the moment of a the force about any point on the line of interest using the cross product, (4.3.4).

$$\mathbf{M} = (\mathbf{r} \times \mathbf{F})$$

- Next, find the scalar projection of the moment onto the line of interest using a dot product, (2.7.10).

$$M_{line} = \|\text{proj}_{\mathbf{u}} \mathbf{M}\| = \mathbf{M} \cdot \hat{\mathbf{u}}$$

Scalar Triple Product: Combines the cross and dot product into the determinant of a 3×3 matrix formed from the three vectors above.

- The scalar triple product consists of taking the determinant of a matrix consisting of the components of the line's unit vector $\hat{\mathbf{u}}$ in the top row, the components of a position vector $\mathbf{r}$ in the middle row, and the components of the $\mathbf{F}$ in the bottom row. The determinant can be computed in multiple ways, but we suggest using the augmented determinant method Figure 2.8.6 or computing it directly in a calculator which allows 3×3 matrix computations.
- Here are the computations:

$$\begin{aligned} M_{line} &= \|\text{proj}_{\mathbf{u}} \mathbf{M}\| = \hat{\mathbf{u}} \cdot (\mathbf{r} \times \mathbf{F}) \\ &= \begin{vmatrix} u_x & u_y & u_z \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix} \\ &= (r_y F_z - r_z F_y) u_x + (r_z F_x - r_x F_z) u_y + (r_x F_y - r_y F_x) u_z \end{aligned}$$

- Note that the above determinant does *not* contain any unit vectors, like the $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ vectors that show up in the top of cross-product determinants. Therefore, you are just multiplying the *numbers* in the diagonals and adding them up to a scalar value.

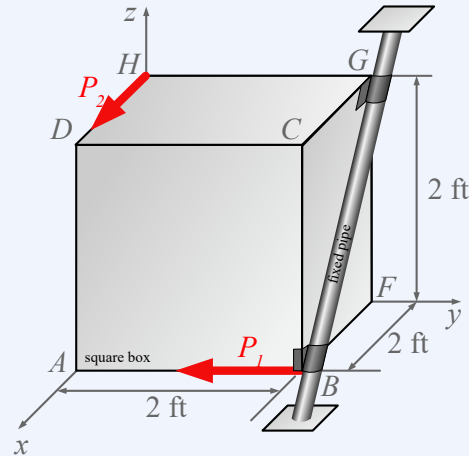
A positive value indicates that the moment vector points in the $\hat{\mathbf{u}}$ direction, while a negative value indicates the moment points in the opposite direction.

Finally if you need to find the vector components of this moment about a line, multiplying the scalar projection by the line's unit vector will add direction to your scalar moment value, (2.7.11)

$$\mathbf{M}_{line} = M_{\hat{\mathbf{u}}} = \|\text{proj}_{\mathbf{u}} \mathbf{M}\| \hat{\mathbf{u}}. \quad (4.4.5)$$

Example 4.4.6 3D Moment about a Line.

A thin plate $OABC$ sits in the xy plane. Cable BD pulls with a tension of 2 kN through a frictionless ring at point D . Find the moment caused by the tension force around point O .



A 2 ft per side square box is attached to fixed pipe BG by two frictionless hinges B and G . There are two forces applied to the box, $\mathbf{P}_1 = 30$ lbf and $\mathbf{P}_2 = 25$ lbf.

For this two part example, first find the rotational moment from $\mathbf{P}_1$ about line BG and then find the moment of and $\mathbf{P}_2$ about line BG .

- (a) Find the magnitude of the moment caused by force $\mathbf{P}_1$ about line BG .

Answer. M_{P_1} about $BG = 0$ ft·lbf

Solution. The key observation to find the moment of $\mathbf{P}_1$ about line BG is that the line of action of $\mathbf{P}_1$ goes *directly* through line BG , thus force $\mathbf{P}_1$ produces no moment about line BG . Force $\mathbf{P}_1$ still pulls on the box and line BG , but causes no rotation *about* line BG . While you could set up a two-step or mixed triple to compute this moment, the simplest moment arm from line BG to force $\mathbf{P}_1$ would be $\mathbf{r} = \langle 0, 0, 0 \rangle$ ft which results in a M_{P_1} about $BG = 0$ ft·lbf.

- (b) Find the magnitude of the moment caused by force $\mathbf{P}_2$ about line BG .

Answer. M_{P_2} about $BG = 35.355$ ft·lbf

Solution. Keep in mind that to compute the magnitude of a moment about a line, we need to compute a vector moment about a point on the line and then dot this vector moment onto the line's unit vector. There are two options for this computation:

Option 1: The Two-Step Method separates the cross and dot products into two distinct steps.

- Compute the vector moment of $\mathbf{P}_1$ about any point on line BG , including $\mathbf{M}_G$ or $\mathbf{M}_B$, then

- Dot $\mathbf{M}$ onto the unit vector $\widehat{BG}$

Option 2: The Scalar Triple Product combines the cross and dot product into a single matrix determinant, including the following vectors:

- Top row: Unit vector for the line of interest $\widehat{BG}$
- Middle row: Moment arm from line of interest to force vector. We'll use $\mathbf{r}_{HG}$ as it has the most zeros, which will simplify the algebra when computing the determinant.
- Bottom row: Components of force vector $\mathbf{P}_2$.

Let's set up the vectors for the scalar triple product, starting with the unit vector $\widehat{BG}$.

The unit vector $\widehat{BG}$ is found by dividing the position vector $\mathbf{BG}$ by its own magnitude.

$$\widehat{BG} = \frac{\mathbf{BG}}{BG} = \frac{\langle -2.0, 0.0, 2.0 \rangle \text{ ft}}{\sqrt{(-2)^2 + 0^2 + 2^2} \text{ ft}} = \langle -0.707, 0.0, 0.707 \rangle$$

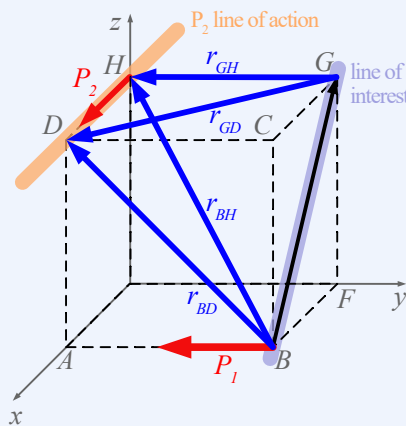
Next, the middle row of the scalar triple product is one of the moment arms from $\mathbf{BG}$ to the force $\mathbf{P}_2$. There are four options that will all result in the same answer after you compute the scalar triple product.

$$\mathbf{r}_{GH} = \langle 0.0, -2.0, 0.0 \rangle \text{ ft}$$

$$\mathbf{r}_{GD} = \langle 2.0, -2.0, 0.0 \rangle \text{ ft}$$

$$\mathbf{r}_{BH} = \langle -2.0, -2.0, 2.0 \rangle \text{ ft}$$

$$\mathbf{r}_{BD} = \langle 0.0, -2.0, 2.0 \rangle \text{ ft}$$



We'll use $\mathbf{r}_{GH}$ going forward as it is a one-dimensional vector and the zeros will simplify the determinant.

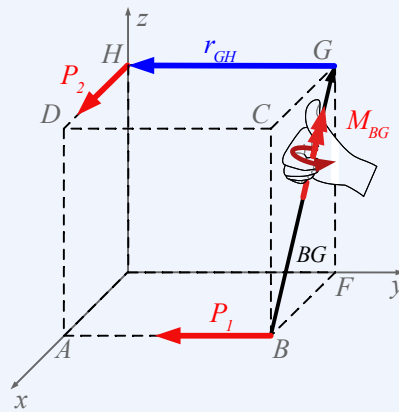
Finally, the bottom row of the scalar triple product are the force vector components.

$$\mathbf{P}_2 = \langle 25, 0, 0 \rangle \text{ lbf}$$

Now, combining the three vectors, here is the triple scalar product matrix:

$$M_{P_2} \text{ about } BG = \begin{vmatrix} -0.707 & 0 & 0.707 \\ 0.0 \text{ ft} & -2.0 \text{ ft} & 0.0 \text{ ft} \\ 25 \text{ lbf} & 0 \text{ lbf} & 0 \text{ lbf} \end{vmatrix} = 35.355 \text{ ft}\cdot\text{lbf}$$

Hence, the magnitude of the moment caused by force $\mathbf{P}_2$ about line $\mathbf{BG}$ is $35.355 \text{ ft}\cdot\text{lbf}$. As we found a positive answer, this confirms that the rotation caused by $\mathbf{P}_2$ is in the same direction as the vector $\mathbf{BG}$, which is reinforced by the right-hand rule thumb direction in the diagram.



Validating the answer with the 2-step approach, we could first sum the moment of $\mathbf{P}_2$ about point G.

$$\mathbf{M}_G = (2 \text{ ft})(25 \text{ lbf}) \hat{k}$$

Given that the $\mathbf{M}_G \hat{k}$ moment and line $\mathbf{BG}$ lie in a 2-D plane, you could even use a 45° right triangle instead of a dot product to find the component of $\mathbf{M}_G$ which lies along line $\mathbf{BG}$.

$$M_{BG} = M_G(\cos 45^\circ)$$

$$M_{BG} = 50 \text{ ft} \cdot \text{lbf}(0.707)$$

$$M_{BG} = 35.355 \text{ ft} \cdot \text{lbf}$$

Not all problems have simple geometry where you can form a right triangle to perform a dot product, but this process demonstrates the two-step combination of a cross and a dot product to find a moment about a line.

4.5 Couples

Key Questions

- What makes a couple different than a typical $\mathbf{r} \times \mathbf{F}$ moment?
- Why is a couple considered a pure moment?
- If a couple is applied about the point we are summing moments, does it still need to be included in the sum of moments equation?

The moments we have considered so far were all caused by single forces producing rotation about a moment center. In this section, we will consider another type of moment, called a **couple**.

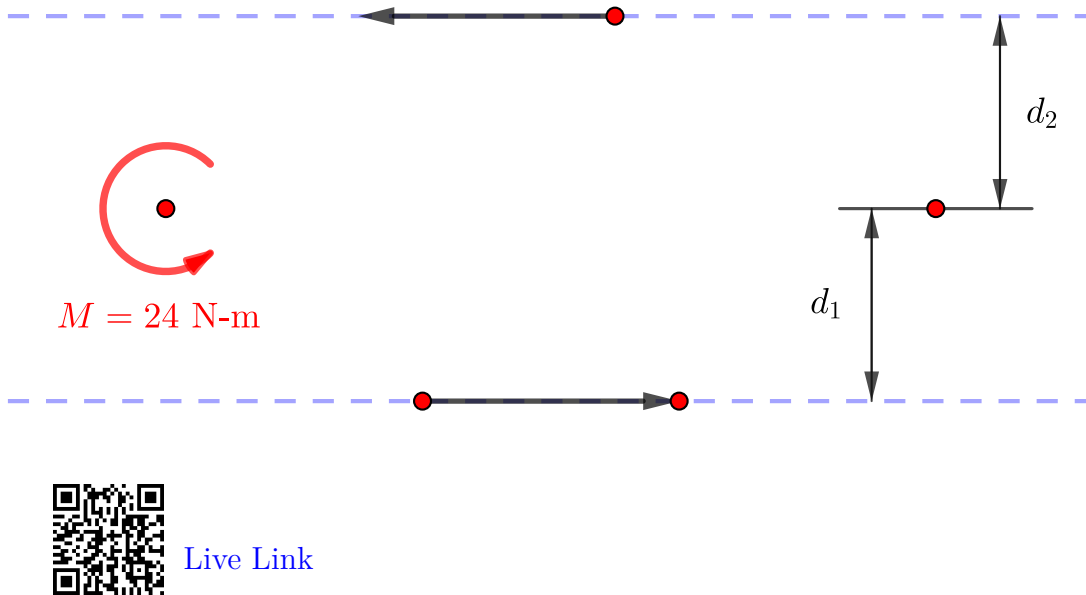
A couple consists of two parallel forces, equal in magnitude, opposite in direction, and non-coincident. Couples are special because the pair of forces always cancel each other, which means that a couple produces a rotational effect but never translation. For this reason, couples are sometimes referred to as “pure moments.” The strength of the rotational effect is called the **moment of the couple** or the **couple-moment**.

When a single force causes a moment about a point, the magnitude depends on the magnitude of the force and the location of the point. In contrast, the moment of a couple is the same at every point and only depends on the magnitude of the opposite forces and the distance between them.

For example, consider the interactive where two equal and opposite forces with different lines of action form a couple. The moment of this couple is found by summing the moments of the two forces about an arbitrary moment center A , applying positive or negative signs for each term according to the right-hand rule. The moment of the couple is always

$$M = Fd_{\perp} \tag{4.5.1}$$

where $d_{\perp}$ is the perpendicular distance between the lines of action of the forces.



[Live Link](#)

Figure 4.5.1 Moment of a couple.

In two dimensions, couples are represented by a curved arrow indicating the direction of the rotational effect. Following the right-hand rule, the value will be positive if the moment is counterclockwise and negative if it is clockwise. In three dimensions, a couple is represented by a normal vector arrow.

When adding moments to find the total or resultant moment, you must include couple-moments as well the $\mathbf{r} \times \mathbf{F}$ moments. In equation form, we could express this as:

$$\Sigma M_P = \Sigma(\mathbf{r} \times \mathbf{F}) + \Sigma(\mathbf{M}_{\text{couple}})$$

Thinking Deeper 4.5.2 Location Independence.

In this section, we have shown that couples produce the same moment at every point on the body. This means that the external effect of couples is *location independent*. Because the moment of a couple is location independent, the moment vector is not bound to any particular point and for this reason is a *free vector*.

We will learn in [Chapter 8](#) that moving a couple around on a rigid body does affect the *internal* loads or stresses inside a body, but changing the location of a couple does not change the *external* loading or reactions.

4.6 Equivalent Transformations

Key Questions

- What is an equivalent transformation?
- What are some examples of equivalent transformations?
- What are *external effects*?

An *equivalent transformation* occurs when a loading on an object is replaced with another loading that has the same *external* effect on the object. By external effect, we mean the response of the body that we can see from the outside, with no consideration of what happens to it internally. If the object is a free body, the external effect would be translation and rotation. In statics, since objects are not accelerating, the external effect really means the reactions at the supports required to maintain equilibrium. The external effects will be exactly the same before and after an equivalent transformation.

Equivalent transformation permits us to swap out one set of forces with another one without changing the fundamental physics of the situation. This is usually done to simplify or clarify the situation, or to give you an alternate way to think about, understand, and solve a mechanics problem.

You already know several equivalent transformations, although we have not used this terminology before. Here are some transformations you applied before.

Vector Addition. When you add forces together using the rules of vector addition, you are performing an equivalent transformation. You can swap out two or more components and replace them with a single equivalent resultant force.

Any number of *concurrent* forces can be added together to produce a single resultant force. By definition, the lines of action of concurrent forces all intersect at a common point. The resultant must be placed at this intersection point for the replacement to be equivalent. This is because before and after the replacement, the moment about the intersection point is zero. If the resultant was placed somewhere else, that would not be true.

Replacing a Force with its Component. Resolving forces into components is also an equivalent transformation, as it is the inverse operation of vector addition. The components are usually orthogonal and in the coordinate directions, or in a given plane and perpendicular to it, but any combination of force components that add to the original force is equivalent.

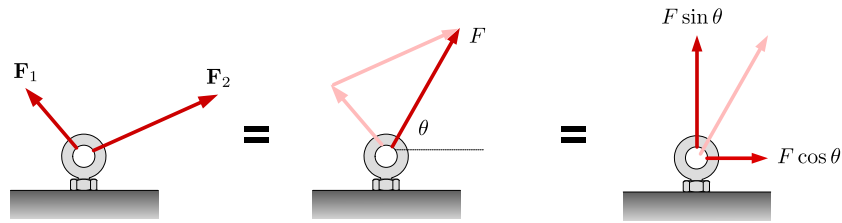


Figure 4.6.1 Equivalent transformations of vectors

In this diagram,

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 = (F; \theta) = \langle F \cos \theta, F \sin \theta \rangle.$$

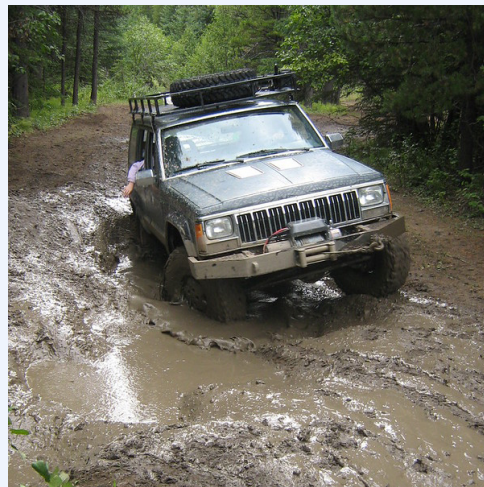
The effects of the force in the x , y , and (in three dimensions the z) directions remain the same, and by Varignon's theorem, we know that the moment these forces make about any point will also be the same.

An interesting special case occurs when two forces are equal and opposite and have the same line of action. When these are added together, they cancel out, so replacing these two forces with nothing is an equivalent transformation. The opposite is true as well, so you can make two equal and opposite forces spontaneously appear at a point if you wish.

Thinking Deeper 4.6.2 Internal Effects.

We made a point of saying that equivalent systems of force have the same *external* effect on the body. This implies that there may be some other effects that are not the same. As you will see in [Chapter 8](#), we sometimes need to consider internal forces and moments. These are the forces inside a body that hold all the parts of the object to each other, otherwise, it would break apart and fail. Although the external effects are the same for all equivalent systems, the internal forces depend on the specifics of how the loads are applied.

Let's imagine that you have gone off-roading and have managed to get your Jeep stuck in the mud. You have two basic options to get it out: you can *pull* it out using the winch on the front bumper, or you can ask your friend to *push* you out with his truck. Both methods (assuming that they apply forces with the same magnitude, direction, and line of action) are statically equivalent, and both will equally move your vehicle forward.



The difference is what might happen to your vehicle. With one method,

there's a danger that you will rip your front bumper off, with the other you might damage your rear bumper. These are the internal effects, and they depend on *where* the equivalent force is applied. These forces are necessary to maintain rigidity and hold the parts of the body together.

Sliding a force along its line of action. Sliding a force along its line of action is an equivalent transformation because sliding a force does not change its magnitude, direction or the perpendicular distance from the line of action to any point, so the moments it creates do not change either. This transformation is called the “Principle of Transmissibility”.

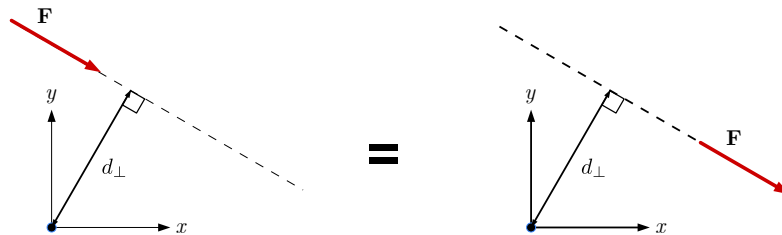


Figure 4.6.3 Sliding a vector along its line of action

Replacing a couple with couple-moment or vice-versa. A *couple*, defined as “two equal and opposite forces with different lines of action,” produces a pure turning action that is equivalent to a concentrated moment, called the *couple-moment*. Couples and couple-moments have no translational effect. Couple-moments are free vectors, which means that they are not bound to any point. Their external effect is on the entire body and is the same regardless of where it is applied.

This means that you are free to swap out a couple for its couple-moment, or swap a couple-moment for a couple that has the same moment, and you may put the replacement anywhere on you please and it will still be equivalent.

The diagram shows a series of equivalent transformations of a couple.

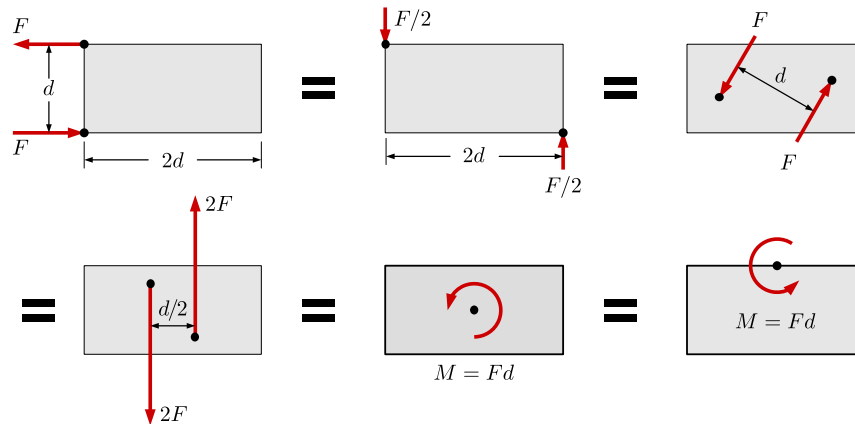


Figure 4.6.4 Equivalent transformations of couples

Concentrated moments are *free vectors*, which you may draw the circular arrow anywhere you like on the body. In other words, moving a concentrated moment from one point to another is an equivalent transformation. Remember, this equivalence only applies to the external effects. What happens inside the body definitely does depend on the specific point where the moment is applied.

Adding moments to produce a resultant moment. If more than one couple-moment or concentrated moment acts on an object, the situation may be simplified by adding them together to produce one *resultant moment*, $\mathbf{M}_R$. The standard rules of vector addition apply.

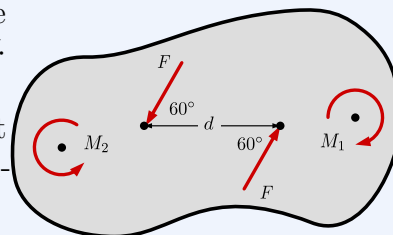
In two-dimensional problems, moments are either clockwise or counterclockwise, so they may be considered scalar values and added algebraically. Give counterclockwise moments a positive sign and clockwise moments a negative sign according to the right-hand rule sign convention. If this is done, the sign of the resultant moment will indicate the direction of the net moment. You can use the right-hand rule to establish the direction of the moment vector, which will point into or out of the page.

$$M_R = \Sigma M$$

Example 4.6.5 Equivalent Moment.

Two concentrated moments and a couple act on the object shown. Given: $M_1 = 400 \text{ N}\cdot\text{m}$, $M_2 = 200 \text{ N}\cdot\text{m}$, $F = 40 \text{ N}$, and $d = 2 \text{ m}$.

Replace these with a single, equivalent concentrated moment, and give the magnitude and direction of your result.



Solution. First, replace the couple with an equivalent couple, M_3 , the magnitude of which is

$$\begin{aligned}
 M_3 &= Fd_{\perp} \\
 &= Fd \sin 60^\circ \\
 &= 69.3 \text{ N}\cdot\text{m}
 \end{aligned}$$

By observation, this is a counterclockwise moment, as is M_2 . M_1 is clockwise. Summing the scalar magnitudes gives the resultant moment. The signs of the terms are assigned according to the sign convention: positive if counterclockwise, negative if clockwise.

$$\begin{aligned}
 M_R &= \Sigma M \\
 &= M_1 + M_2 + M_3 \\
 &= -400 \text{ N}\cdot\text{m} + 200 \text{ N}\cdot\text{m} + 69.3 \text{ N}\cdot\text{m} \\
 &= -130.7 \text{ N}\cdot\text{m} \\
 \mathbf{M}_R &= 130.7 \text{ N}\cdot\text{m} \text{ clockwise}
 \end{aligned}$$

Resolving a moment into components. For three-dimensional moment vectors, another potential equivalent transformation is to resolve a moment vector into components. These may be orthogonal components in the x , y , and z directions, or components in a plane and perpendicular to it, or components in some other rotated coordinate system.

4.7 Statically Equivalent Systems

Key Questions

- What is an equivalent system?
- What is a resultant force?
- What is a resultant moment?
- Do you have to include both $\mathbf{r} \times \mathbf{F}$ moments and couples to find the resultant moment?
- How can you find the simplest equivalent system?
- When will the simplest equivalent system be a wrench?
- How can you determine if two loading systems are statically equivalent?

A loading system is a combination of load forces and moments that act on an object. It can be as simple as a single force, or as complex as a three-dimensional combination of many force and moment vectors.

You will see that any loading systems may be replaced with a simpler *statically equivalent system* consisting of one *resultant force* at a specific point and one *resultant moment* by performing a series of equivalent transformations. Force system resultants provide a convenient representation for complex force interactions at engineering connections that we will rely on later in a variety of contexts. For now, we will focus on the details of reducing a system to a single force and couple.

Depending on the original loading system, the resultant force, the resultant moment, or both may be zero. If they are both zero, it indicates that the object is in equilibrium under this load condition. If they are non-zero, the supports will need to provide an equal and opposite reaction to put the object into equilibrium.

The resultant force acting on a system, $\mathbf{R}$, can be found from adding the individual forces, $\mathbf{F}_i$, such that

$$\mathbf{R} = \sum \mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots$$

The resultant moment, $\mathbf{M}_O$, about a point O , can be found from adding all of the moments $\mathbf{M}$, about that point, including both $\mathbf{r} \times \mathbf{F}$ moments and concentrated moments.

$$\mathbf{M}_O = \sum \mathbf{M}_i = \mathbf{M}_1 + \mathbf{M}_2 + \mathbf{M}_3 + \dots$$

It is often more convenient to work with the scalar components of the resultant vectors since they separate the effects in the three coordinate directions.

$$\begin{array}{ll} R_x = \Sigma F_x & M_{Ox} = \Sigma M_x \\ R_y = \Sigma F_y & M_{Oy} = \Sigma M_y \\ R_z = \Sigma F_z & M_{Oz} = \Sigma M_z \end{array}$$

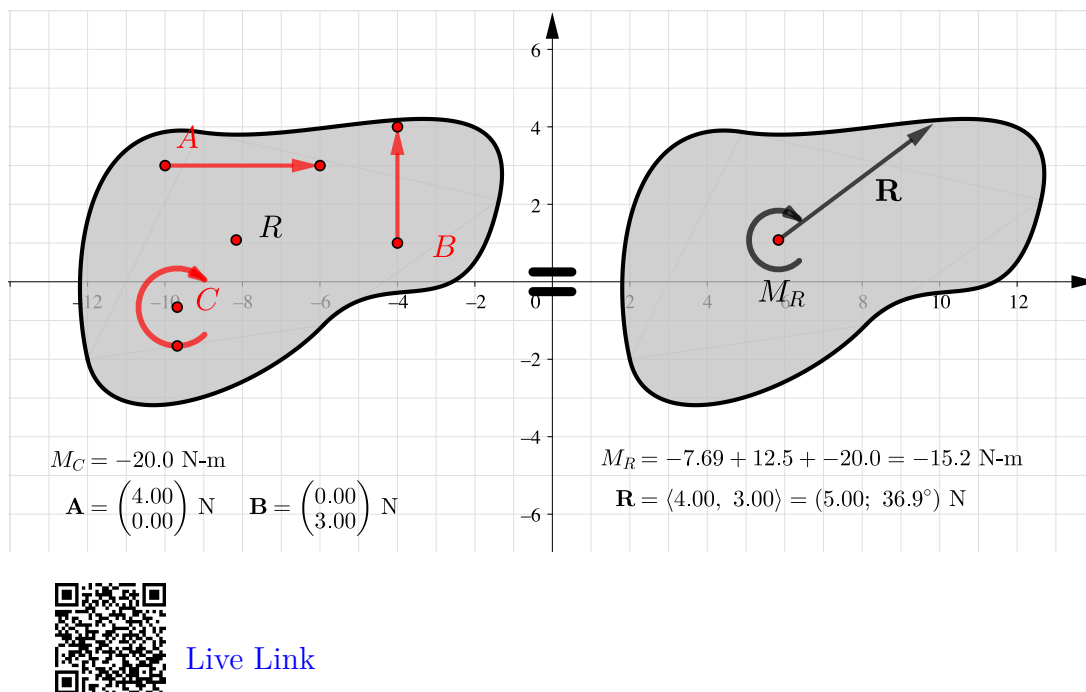


Figure 4.7.1 Statically equivalent systems

Force-Couple Systems. One transformation you might want to make is to move a force to another location. While *sliding* a force along its line of action is fine, *moving* a force to another point changes its line of action and thus its rotational effect on the object, so moving a force to a new line of action is *not* an equivalent transformation.

Consider the cantilever beam below. In diagram (a), the load P is at the end of the beam, and in (b) it has been moved to the center. The external effects are shown in (c) and (d). Although the vertical reaction force is the P in both cases, the reaction moment at point O is $2P\ell$ in the first case and $P\ell$ in the second.

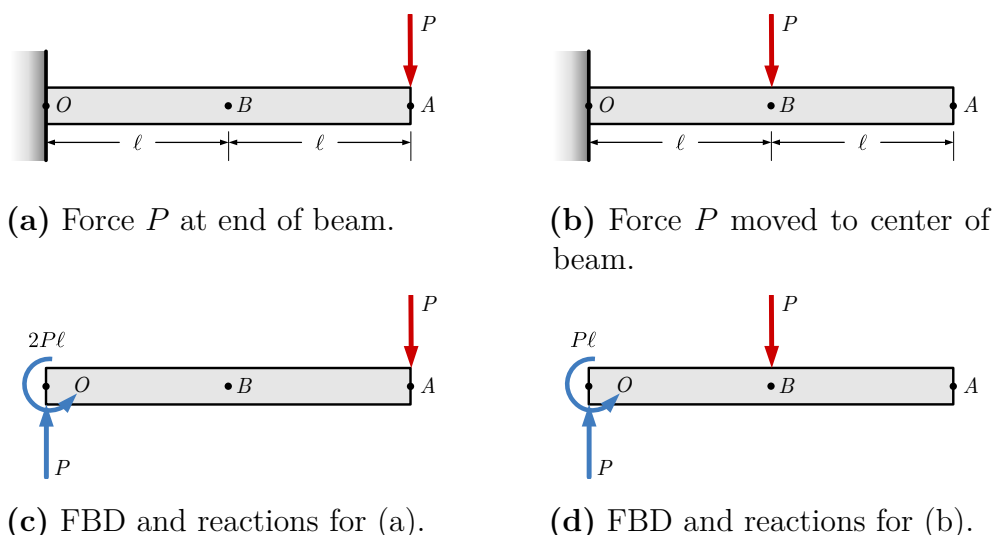


Figure 4.7.2 Moving a force is not an equivalent transformation

You *can* move a force to a new line of action in an equivalent fashion if you add a “compensatory couple” to undo the effect of changing the line of action. This can be accomplished with a series of individual equivalent transformations as shown in the diagram below. To move P to another location, first add two equal and opposite forces where you want the force to be, as in (b). Then recognize the couple you have formed (c), and replace it with an equivalent couple-moment. The result of this process is the *equivalent force-couple system* shown in diagram (d), which is statically equivalent to the original situation in (a).

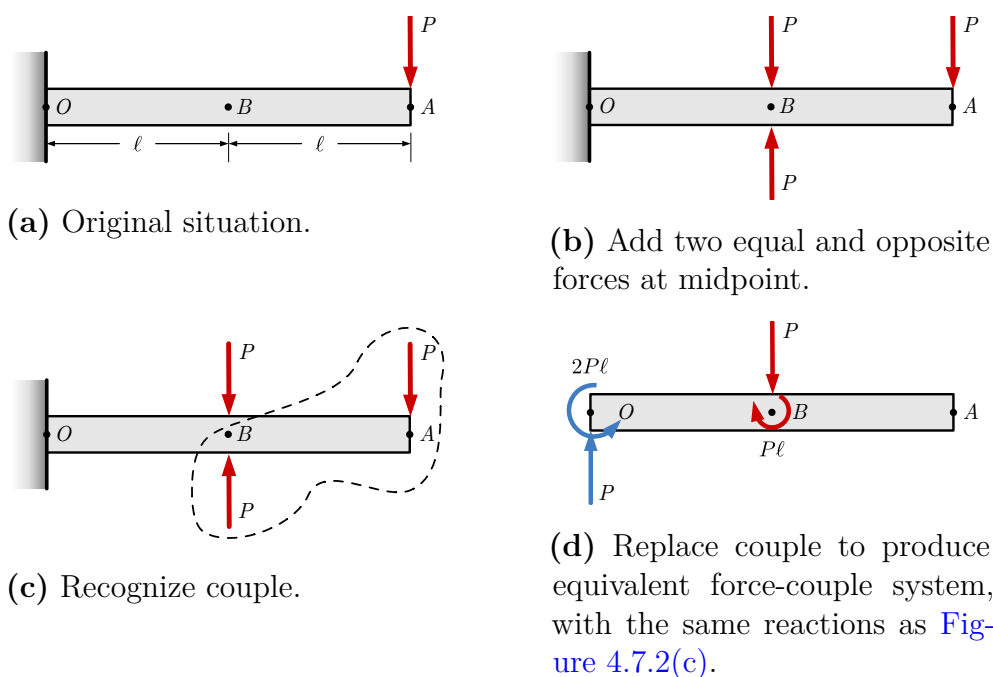


Figure 4.7.3 Equivalent Force-couple system

Evaluating the moment at point O was an arbitrary choice. Any other point would give the same result. For example, in the original situation (a) force P makes a clockwise moment $M = P\ell$ about the midpoint. When the force is moved to the center P creates no moment there, so a clockwise compensatory couple with a magnitude of $P\ell$ must be added to maintain equivalence. This is the same result as we found previously (d). The compensatory couple has been drawn centered around the midpoint, but this too is arbitrary because concentrated moments are free vectors and can be placed at any location.

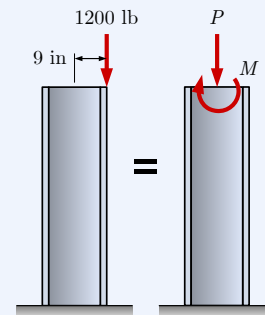
Reduction of a complex system. Any loading system can be reduced to a statically equivalent system consisting of single force and a single moment at a specified point with the following procedure:

1. Determine the resultant moment about the specified point by considering all forces and concentrated moments on the original system.
2. Determine the resultant force by adding all forces acting on the original system.
3. Determine the resultant moment about a point in the original system
4. Create the statically equivalent system by replacing all loads with the resultant force and the resultant moment at the selected point.

Example 4.7.4 Eccentric loading.

An vertical column is supporting an eccentric load as shown.

Replace this load with an equivalent force-couple system acting at the center of the beam's top surface.



Solution. In order to move the vertical force 9 in to the left, a clockwise couple M must be added to maintain equivalence, where

$$\begin{aligned}
 M &= Pd \\
 &= (1200 \text{ lb})(9 \text{ in}) \\
 &= 10800 \text{ in}\cdot\text{lb} \\
 &= 900 \text{ ft}\cdot\text{lb}
 \end{aligned}$$

Example 4.7.5 Equivalent Force-couple System.

- Replace the system of forces in diagram (a) with an equivalent force-couple system at point A .
- Replace this force-couple system with a single equivalent force and specify its location.

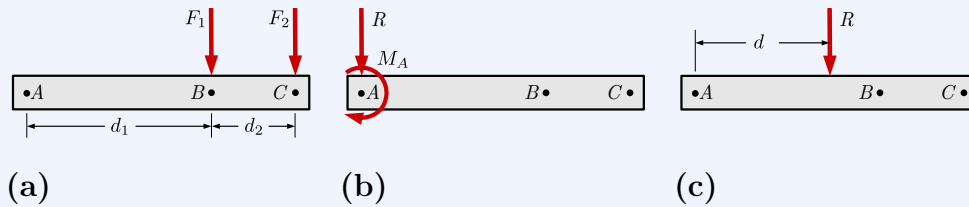


Figure 4.7.6

Solution. The original system is shown in (a). Since the F_1 and F_2 are parallel, the magnitude of the resultant force is just the sum of the two magnitudes and it points down.

$$R = F_1 + F_2$$

The resultant moment about point A is

$$M_A = F_1 d_1 + F_2 (d_1 + d_2).$$

To create the equivalent system (b), the resultant force and resultant moment are placed at point A .

The system in (b) can be further simplified to eliminate the moment at M_A , by performing the process in reverse.

In (c) we place the resultant force R a distance d away from point A such that the resultant moment around point A remains the same. This distance can be found using $M = Fd$.

$$d = M_A / R$$

The systems in (a), (b), and (c) are all statically equivalent

In this example, we started with two forces. We have found two different statically equivalent systems; one with a force and a couple, the other with a single force. This latter system is simpler than the original system.

It is important to note that static equivalence applies to external effects only. When determining internal forces, such as the shear and bending moment discussed in [Section 8.4](#) or when considering non-rigid bodies, the original loading system must be used.

Determining Equivalence. Two complex loading systems are equivalent if they reduce to the same resultant force and the same resultant moment about any arbitrary point.

Two loading systems are statically equivalent if

- The resultant forces are the equal
- The resultant moments about some point are equal

This process is illustrated in the following example.

Example 4.7.7 Finding Statically Equivalent Loads.

Which of the three loading systems shown are statically equivalent?

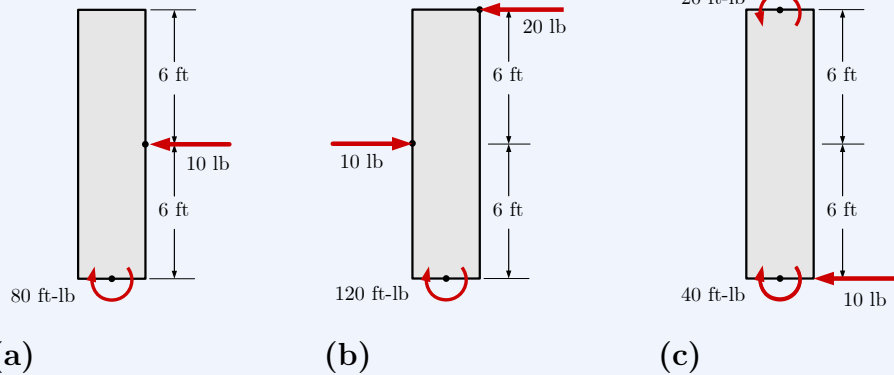


Figure 4.7.8

Solution.

1. *Strategy.*

Evaluate the resultant force and resultant moment for each case and compare. We choose to evaluate the resultant moment about point A, though any other point would work.

2. *For system (a).*

$$\begin{aligned}\mathbf{R} &= \langle -10, 0 \rangle \text{ lb} \\ \mathbf{M}_A &= -80 + 6(10) \\ &= -20 \text{ ft}\cdot\text{lb}\end{aligned}$$

3. *For system (b).*

$$\begin{aligned}\mathbf{R} &= \langle -20 + 10, 0 \rangle \text{ lb} \\ &= \langle -10, 0 \rangle \text{ lb}\end{aligned}$$

$$\begin{aligned}\mathbf{M}_A &= -120 + 12(20) - 6(10) \\ &= 60 \text{ ft}\cdot\text{lb}\end{aligned}$$

4. For system (c).

$$\begin{aligned}\mathbf{R} &= \langle -10, 0 \rangle \text{ lb} \\ \mathbf{M}_A &= -40 + 20 + 0(10) \\ &= -20 \text{ ft}\cdot\text{lb}\end{aligned}$$

Systems (a) and (c) are statically equivalent since $\mathbf{R}$ and $\mathbf{M}_A$ are the same in both cases. System (b) is not as its resultant moment is different than the other two.

Any load system can be simplified to its resultant force $\mathbf{R}$, and resultant couple $\mathbf{M}$, acting at any arbitrary point O . There are four common special cases, which are worth highlighting individually.

Concurrent forces. When all forces in a system are concurrent, the resultant moment about their common intersection point will always be zero. We then need only find the resultant force and place it at the point of intersection. The resultant moment about any other point is the moment of the resultant force $\mathbf{R}$ about that point.

Parallel forces. When all forces in a system are parallel, the resultant force will act in this direction with a magnitude equal to the sum of the individual magnitudes. There will be no moment created about this axis, but we need to find the resultant moment about the other two rectangular axes. That is, if all forces act in the x direction, we need only find the resultant force in the x direction and the resultant moment about the y and z axes.

Coplanar forces. When all forces in a system are coplanar we need only find the resultant force in this plane and the resultant moment about the axis perpendicular to this plane. That is, if all forces exist in the x - y plane, we need only to sum components in the x and y directions to find resultant force $\mathbf{R}$, and use these to determine the resultant moment about the z axis. All two-dimensional problems fall into this category.

Wrench resultant. A wrench resultant is a special case where the resultant moment acts around the axis of the resultant force. The directions of the resultant force vector and the resultant moment vector are the same.

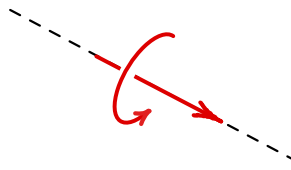


Figure 4.7.9 Wrench Resultant

For example, if the resultant force is only in the x direction and the resultant moment acts only around the x axis, this is an example of a wrench resultant. An everyday example is a screwdriver, where both the resultant force and axis of rotation are in-line with the screwdriver. A wrench resultant is considered positive if the couple vector and force vector point in the same direction, and negative if they point in opposite directions.

Any three-dimensional force-couple system may be reduced to an equivalent wrench resultant even if the resultant force and resultant moment do not initially form a wrench resultant.

To find the equivalent wrench resultant:

1. First, find the resultant force $\mathbf{R}$ and resultant moment $\mathbf{M}$ at an arbitrary point, O . These need not act along the same axis.
2. Resolve the resultant moment into scalar components $M_{\parallel}$ and $M_{\perp}$, parallel and perpendicular to the axis of the resultant force.
3. Eliminate $M_{\perp}$ by moving the resultant force away from point O by distance $d = M_{\perp}/R$

The simplified system consists of moment $\mathbf{M}_{\parallel}$ and force $\mathbf{R}$ and acting distance d away from point O . Since $\mathbf{R}$ and $\mathbf{M}_{\parallel}$ act along the same axis, the system has been reduced to a wrench resultant. Wrench resultants are the most general way to represent a complex force-couple system, but their utility is limited.

4.8 Exercises (Ch. 4)

Chapter 4 Exercises

Moments		0/165
Definition of a Moment	0/30	Not attempted
Varignon's Theorem	0/50	Not attempted
Moment of a Force about a Point	0/20	Not attempted
Semicircle	0/15	Not attempted
Opposing Moments using Varignon's Theorem	0/20	Not attempted
Moment Balance: L shape	0/10	Not attempted
Moment Balance: Z shape	0/20	Not attempted
Couples		0/40
Moment of a Couple	0/15	Not attempted
Resultant Couple	0/15	Not attempted
Couples in Equilibrium	0/10	Not attempted
Equivalent Systems		0/115
Two Parallel Forces	0/20	Not attempted
Equivalent Force-Couple System: Circle	0/30	Not attempted
Equivalent Force-Couple System: Bracket	0/35	Not attempted
Resultant of a Force-Couple System	0/30	Not attempted

Display options

Done

Created using [Numbas](#) v10.0, developed by [Newcastle University](#).



Standalone
In Context

Chapter 5

Rigid Body Equilibrium

This chapter will investigate the equilibrium of simple rigid bodies like your book, phone, or pencil. The important difference between rigid bodies and the particles of [Chapter 3](#) is that rigid bodies have the potential to rotate around a point or axis, while particles do not.

For rigid body equilibrium, we need to maintain translational equilibrium with

$$\sum \mathbf{F} = 0 \quad (5.0.1)$$

and also maintain a balance of rotational forces and couple-moments with a new equilibrium equation

$$\sum \mathbf{M} = 0. \quad (5.0.2)$$

5.1 Degree of Freedom

Degrees of freedom refers to the number of independent parameters or values required to specify the *state* of an object.

The *state* of a *particle* is completely specified by its location in space, while the state of a rigid body includes its location in space and also its orientation.

Two-dimensional rigid bodies in the xy plane have three degrees of freedom. Position can be characterized by the x and y coordinates of a point on the object and orientation by angle θ_z about an axis perpendicular to the plane. The complete movement of the body can be defined by two linear displacements Δx and Δy , and one angular displacement $\Delta\theta_z$.

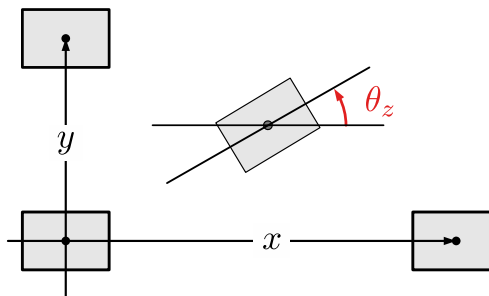


Figure 5.1.1 Two-dimensional rigid bodies have three degrees of freedom.

Three-dimensional rigid bodies have six degrees of freedom, which can be specified with three orthogonal coordinates x, y and z , and three angles of rotation, θ_x, θ_y and θ_z . Movement of the body is defined by three translations $\Delta x, \Delta y$ and Δz , and three rotations $\Delta\theta_x, \Delta\theta_y$ and $\Delta\theta_z$.

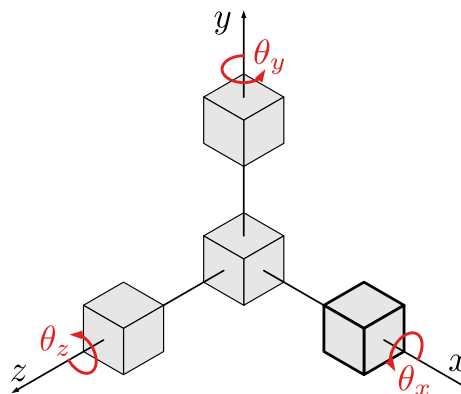


Figure 5.1.2 Three-dimensional rigid body have six degrees of freedom - three translations and three rotations.

For a body to be in static equilibrium, all possible movements must be adequately restrained. If a degree of freedom is not restrained, the body is in an unstable state, free to move in one or more ways. Stability is highly desirable for reasons of human safety, and bodies are often restrained by *redundant* restraints so that if one were to fail, the body would still remain stable. If the restraints correctly interpreted, then equal constraints and degrees of freedom create a stable system, and the values of the reaction forces and moments can be determined using equilibrium equations. If the number of restraints exceeds the number of degrees of freedom, the body is in equilibrium but you will need techniques we won't cover in statics to determine the reactions.

5.2 Free-Body Diagrams

Key Questions

- What are the five steps to create a free-body diagram?
- What are degrees of freedom, and how do they relate to stability?
- Which reaction forces and couple-moments come from each support type?
- What are the typical support force components and couple-moment components that can be modeled from the various types of supports?

Free body diagrams are the tool that engineers use to identify the forces and moments that influence an object. They will be used extensively in statics, and you will use them again in other engineering courses so your effort to master them now is worthwhile. Although the concept is simple, students often need help to draw them correctly.

Drawing a correct free-body diagram is the first and most important step

in the process of solving an equilibrium problem. It is the basis for all the equilibrium equations you will write; if your free-body diagram is incorrect, your equations, analysis, and solutions will also be wrong.

A quality free-body diagram is neat, clearly drawn, and contains all the information necessary to solve the equilibrium. You should take your time and think carefully about the free-body diagram before you begin to write and solve equations. A straightedge, protractor and colored pencils all can help. You will inevitably make mistakes that will lead to confusion or incorrect answers; you are encouraged to think about these errors and identify any misunderstandings to avoid them in the future.

Every equilibrium problem begins by drawing and labeling a free-body diagram!

Creating Free Body Diagrams. The basic process for drawing a free-body diagrams is

1. *Select and isolate an object.*

The “free-body” in free-body diagram means that the body to be analyzed must be free from the supports that are physically holding it in place.

Simply sketch a quick outline of the object as if it is floating in space disconnected from everything. Do *not* draw free-body diagram forces on top of your problem drawing — the body needs to be drawn free of its supports.

2. *Select a reference frame.*

Select a right-handed coordinate system to use as a reference for your equilibrium equations. Even if you are using a horizontal x axis and vertical y axis, indicate your coordinate system on your diagram.

Look ahead and select a coordinate system that minimizes the number of unknown force components in your equations. The choice is technically arbitrary, but a good choice will simplify your calculations and reduce your effort. If you and another student pick different reference systems, you should both get the same answer while expressing your work with different components.

3. *Identify all loads.*

Add vector arrows representing the applied forces and couple moments of acting on the body. These are often obvious. Include the body’s weight if it is non-negligible. If a vector has a known line of action, draw the arrow in that direction; if its sense is unknown, assume one. Every vector should have a descriptive variable name and a clear arrowhead indicating its direction.

4. *Identify all reactions.*

Traverse the perimeter of the object and wherever a support was removed when isolating the body, replace it with the forces and/or couple-moments which it provides. Label each reaction with a descriptive variable name and a clear arrowhead. Again, if a vector's direction is unknown just assume one.

The reaction forces and moments provided by common two-dimensional supports are shown in [Figure 5.2.1](#) and three dimensional support in [Figure 5.2.2](#). Identifying the correct reaction forces and couple-moments coming from supports is perhaps the most challenging step in the entire equilibrium process.

5. *Label the diagram.*

Verify that every dimension, angle, force, and moment is labeled with either a value or a symbolic name if the value is unknown. Supply the information needed for your calculations, but don't clutter the diagram up with unneeded information. This diagram should be a stand-alone presentation.

Drawing good free-body diagrams is surprisingly tricky and requires practice. Study the examples, think hard about them, do lots of problems, and learn from your errors.

Two-dimensional Reactions. Supports supply reaction forces and moment which prevent bodies from moving when loaded. In the most basic terms, forces prevent translation, and moments prevent rotation.

The reactions supplied by a support depend on the nature of the particular support. For example in a top view, a door hinge allows the door to rotate freely but prevents it from translating. We model this as a frictionless pin that supplies a perpendicular pair of reaction forces, but no reaction moment. We can evaluate all the other physical supports in a similar way to come up with the table below. You will notice that some two-dimensional supports only restrain one degree of freedom and others restrain up to three degrees of freedom. The number of degrees of freedom directly correlates to the number of unknowns created by the support.

The table below shows typical two-dimensional support methods and the corresponding reaction forces and moments supplied each.

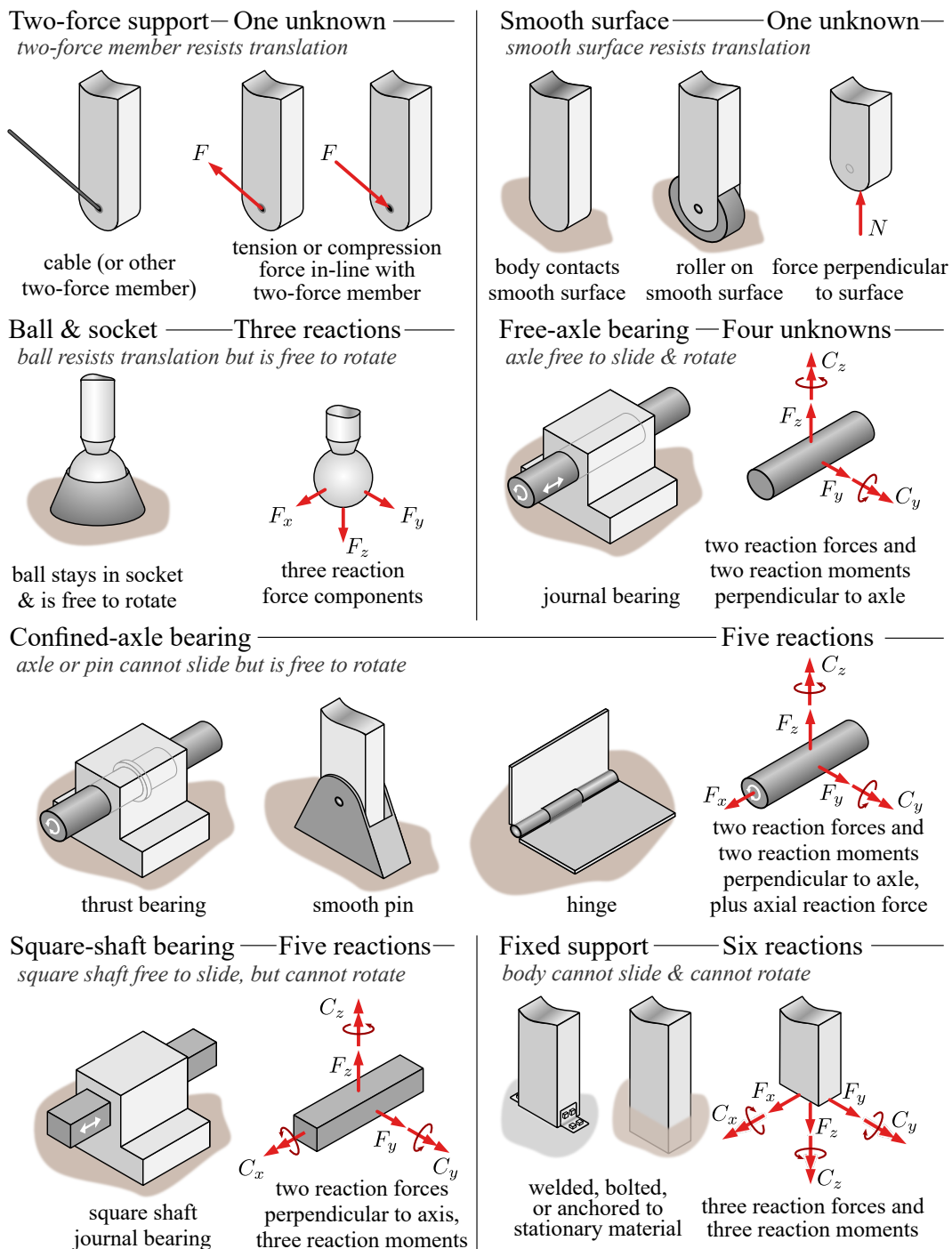


Figure 5.2.2 Table of common three-dimensional supports and their associated reactions.

One new issue we face in three-dimensional problems is that reaction couples may be **available** but not **engaged**.

A support which provides a non-zero reaction is said to be engaged. Picture a crate sitting at rest on a horizontal surface with a cable attached to the top of the crate. If the cable is slack, the reaction of the cable would be available but not *engaged*. Instead, the floor would be supporting the full weight of the

crate. If we were to remove the floor, the cable would be engaged and support the weight of the crate.

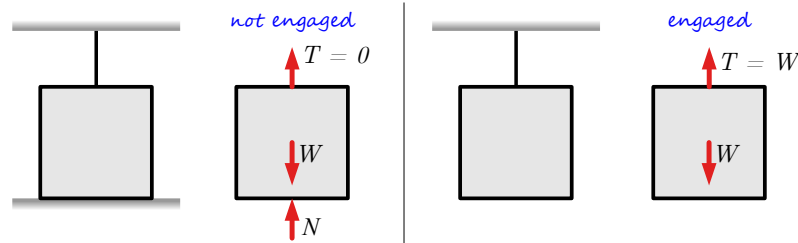


Figure 5.2.3 Available and Engaged reactions.

To get a feel for how reaction couples engage, pick up your laptop or a heavy book and hold it horizontally with your left hand. Can you feel your hand supplying an upward force to support the weight *and* a counter-clockwise reaction couple to keep it horizontal? Now add a similar support by gripping with your right hand. How do the forces and couple-moments change? You should have felt the force of your left hand decrease as your right hand picked up half the weight, and also noticed that the reaction couple from your left hand was no longer needed.

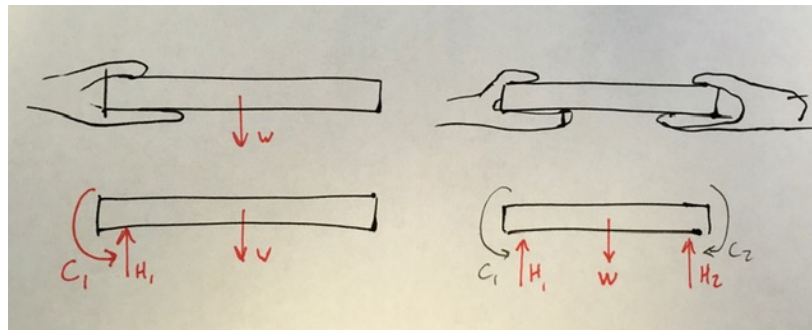


Figure 5.2.4 One hand holding an object versus two hands holding the same object.

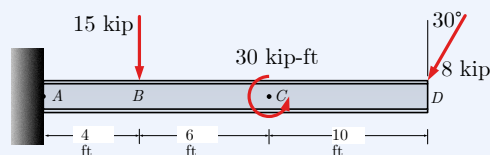
The vertical force in your right hand engaged instead of the couple-moment of your left hand. The reaction couples from both hands are available, but the vertical forces engage first and are sufficient for equilibrium. This phenomena is described by the saying “reaction forces engage before reaction couple-moments”.

Free-Body Diagram Examples. Given that there several options for representing reaction forces and couple-moments from a support, there are different, equally valid options for drawing free-body diagrams. With experience you will learn which representation to choose to simplify the equilibrium calculations.

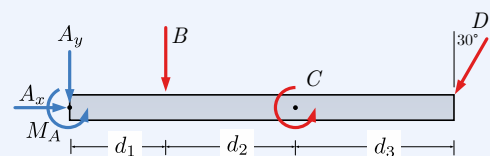
Possible free-body diagrams for two common situations are shown in the next two examples.

Example 5.2.5 Fixed support.

The cantilevered beam is embedded into a fixed vertical wall at A . Draw a neat, labeled, correct free-body diagram of the beam and identify the knowns and the unknowns.

**Solution.**

Begin by drawing a neat rectangle to represent the beam disconnected from its supports, then add all the known forces and couple-moments. Label the magnitudes of the loads and the known dimensions symbolically.



Choose the standard xy coordinate system, since it aligns well with the forces.

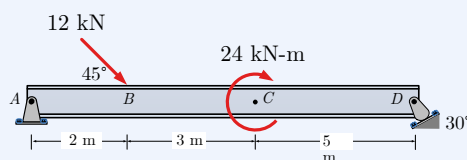
The wall at A is a fixed support which prevents the beam from translating up, down, left or right, or rotating in the plane of the page. These constraints are represented by two perpendicular forces and a concentrated moment, as shown in Figure 5.2.1. Label these unknowns as well.

The knowns in this problem are the magnitudes and directions of moment $\mathbf{C}$, forces $\mathbf{B}$, and $\mathbf{D}$ and the dimensions of the beam. The unknowns are the two force components A_x and A_y and the scalar moment M_A caused by the fixed connection. If you prefer, you may represent force $\mathbf{A}$ as a force of unknown magnitude acting at an unknown direction. Whether you represent it as x and y components or as a magnitude and direction, there are two unknowns associated with force $\mathbf{A}$.

The three unknown reactions can be found using the three independent equations of equilibrium we will discuss later in this chapter.

Example 5.2.6 Frictionless pin and roller.

The beam is supported by a frictionless pin at A and a rocker at D . Draw a neat, labeled, correct free-body diagram of the beam and identify the knowns and the unknowns.

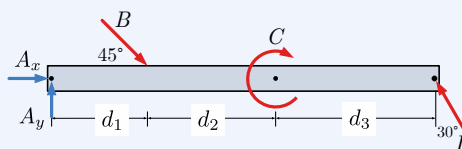


Solution. In this problem, the knowns are the magnitude and direction of force $\mathbf{B}$ and moment $\mathbf{C}$ and the dimensions of the beam.

The constraints are the frictionless pin at A and the rocker at D . The pin prevents translation but not rotation, which means it has two

unknowns, represented by either magnitude and direction, or by two orthogonal components. The rocker provides a force perpendicular to the surface it rests on, which is 30° from the horizontal. This means that the line of action of force **D** is 30° from the vertical, giving us its direction but not its sense or magnitude

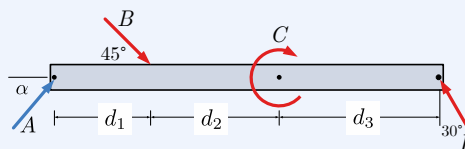
To draw the free-body diagram, start with a neat rectangle to represent the beam disconnected from its supports, then draw and label known force **B** and moment **C** and the dimensions.



Add forces A_x and A_y representing vector **A** and force **D** at **D**, acting 30° from the vertical.

When a force has a known line of action as with force **D**, draw it acting along that line; don't break it into components. When it is not obvious which way a reaction force actually points along its lines of action, just make your best guess and place an arrowhead accordingly. Your calculations will confirm or refute your guess later.

As in the previous example, you could alternately represent force **A** as an unknown magnitude acting in an unknown direction, though there is no particular advantage to doing so in this case.



5.3 Equations of Equilibrium

Key Questions

- What is the definition of static equilibrium?
- How do I choose which are the most efficient equations to solve two-dimensional equilibrium problems?

In statics, our focus is on systems where both linear acceleration **a** and angular acceleration α are zero. These systems are frequently stationary, but could be moving with constant velocity.

Under these conditions [Newton's Second Law for translation](#) reduces to

$$\sum \mathbf{F} = 0, \quad (5.3.1)$$

and, [Newton's second law for rotation](#) gives the similar equation

$$\sum \mathbf{M} = 0. \quad (5.3.2)$$

The first of these equations requires that all forces acting on an object balance and cancel each other out, and the second requires that all moments balance as well. Together, these two equations are the mathematical basis of this course and are sufficient to evaluate equilibrium for systems with up to six degrees of freedom.

These are vector equations; hidden within each are three independent scalar equations, one for each coordinate direction.

$$\sum \mathbf{F} = 0 \implies \begin{cases} \Sigma F_x = 0 \\ \Sigma F_y = 0 \\ \Sigma F_z = 0 \end{cases} \quad \sum \mathbf{M} = 0 \implies \begin{cases} \Sigma M_x = 0 \\ \Sigma M_y = 0 \\ \Sigma M_z = 0 \end{cases} \quad (5.3.3)$$

Working with these scalar equations is often easier than using their vector equivalents, particularly in two-dimensional problems.

In many cases we do not need all six equations. We saw in [Chapter 3](#) that particle equilibrium problems can be solved using the force equilibrium equation alone, because particles have, at most, three degrees of freedom and are not subject to any rotation.

To analyze rigid bodies, which can rotate as well as translate, the moment equations are needed to address the additional degrees of freedom. Two-dimensional rigid bodies have only one degree of rotational freedom, so they can be solved using just one moment equilibrium equation, but to solve three-dimensional rigid bodies, which have six degrees of freedom, all three moment equations and all three force equations are required.

5.4 2D Rigid Body Equilibrium

Two-dimensional rigid bodies have three degrees of freedom, so they only require three independent equilibrium equations to solve. The six scalar equations of [\(5.3.3\)](#) can easily be reduced to three by eliminating the equations which refer to the unused z dimension. For objects in the xy plane there are no forces acting in the z direction to create moments about the x or y axes, so the reduced set of three equations is

$$\{1\} = \begin{cases} \Sigma F_x = 0 \\ \Sigma F_y = 0 \\ \Sigma M_A = 0 \end{cases}$$

where the subscript z has been replaced with a letter to indicate an arbitrary moment center in the xy plane instead of a perpendicular z axis.

This is not the only possible set of equilibrium equations. Either force equation can be replaced with a linearly independent moment equation about a point

of your choosing ¹, so the other possible sets are

$$\{2\} = \begin{cases} \sum F_x &= 0 \\ \sum M_B &= 0 \\ \sum M_A &= 0 \end{cases} \quad \{3\} = \begin{cases} \sum M_C &= 0 \\ \sum F_y &= 0 \\ \sum M_A &= 0 \end{cases} \quad \{4\} = \begin{cases} \sum M_C &= 0 \\ \sum M_B &= 0 \\ \sum M_A &= 0 \end{cases}$$

For set four, moment centers A , B , and C must form a triangle to ensure the three equations are linearly independent.

You have a lot of flexibility when solving rigid-body equilibrium problems. In addition to choosing which set of equations to use, you are also free to rotate the coordinate system to any orientation you like, pick different points for moment centers, and solve the equations in any order or simultaneously.

This freedom raises several questions. Which equation set should you choose? Is one choice ‘better’ than another? Why bother rotating coordinate systems? How do you select moment centers? Students want to know “how to solve the problem,” when in reality there are many ways to do it.

The actual task is to choose an efficient approach and carry it out. An efficient solution is one which avoids mathematical complications and makes the problem easy to solve. Complications include unpleasant geometries, unnecessary algebra, and particularly simultaneous equations, which are algebra intensive and error prone.

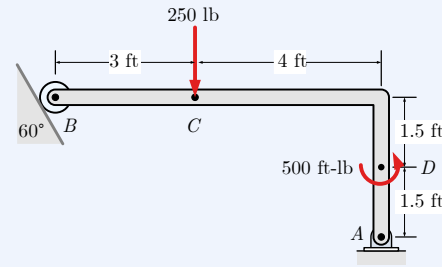
So how do you set up an efficient approach? First, stop, think, and look for opportunities to make the solution more efficient. Here are some recommendations.

1. Equation set one is usually a good choice and should be considered first.
2. Inspect your free-body diagram and identify the unknown values in the problem. These may be magnitudes, directions, angles or dimensions.
3. Align the coordinate system with at least one unknown force.
4. Take moments about the point where the lines of action of two unknown forces intersect, which eliminates them from the equation.
5. Solve equations with one unknown first.

¹Labels A , B and C in these equations are representative. They don’t have to correspond to points A , B and C on your problem.

Example 5.4.1 Pin and Roller.

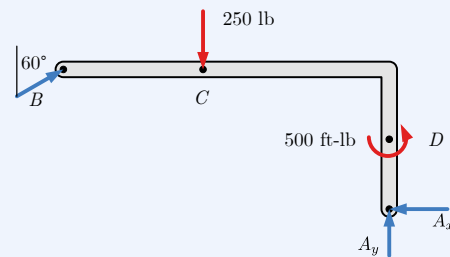
The L-shaped body is supported by a roller at B and a frictionless pin at A . The body supports a 250 lb vertical force at C and a 500 ft·lb couple-moment at D . Determine the reactions at A and B .



This problem will be solved three different ways to demonstrate the advantages and disadvantages of different approaches.

Solution 1.

Solutions always start with a free-body diagram, showing all forces and moments acting on the object. Here, the known loads $C = 250$ lb (down) and $D = 500$ ft·lb (CCW) are red, and the unknown reactions A_x , A_y and B are blue.



The force at B is drawn along its known line-of-action perpendicular to the roller surface, and drawn pointing up and right because that will oppose the rotation of the frame about A caused by load C and moment D . The force at A is represented by unknown components A_x and A_y . The sense of these components is unknown, so we have arbitrarily assigned the arrowheads pointing left and up.

We have chosen the standard coordinate system with positive x to the right and positive y pointing up, and resolved force A into components in the x and y directions.

The magnitude of force B is unknown but its direction is known, so the x and y components of B can be expressed as

$$B_x = B \sin 60^\circ \qquad B_y = B \cos 60^\circ$$

We choose to solve equation set $\{A\}$, and choose to take moments about point A , because unknowns A_x and A_y intersect there. Substituting the variables into the equation and solving for the unknowns gives

$$\begin{aligned} \sum F_x &= 0 \\ B_x - A_x &= 0 \\ A_x &= B \sin 60^\circ \end{aligned} \qquad (1)$$

$$\begin{aligned}
 \sum F_y &= 0 \\
 B_y - C + A_y &= 0 \\
 A_y &= C - B \cos 60^\circ
 \end{aligned} \tag{2}$$

$$\begin{aligned}
 \sum M_A &= 0 \\
 -B_x(3) - B_y(7) + C(4) + D &= 0 \\
 3B \cos 60^\circ + 7B \sin 60^\circ &= 4C + D \\
 B(3 \sin 60^\circ + 7 \cos 60^\circ) &= 4C + D \\
 B &= \frac{4C + D}{6.098}
 \end{aligned} \tag{3}$$

Of these three equations only the third can be evaluated immediately, because we know C and D . In equations (1) and (2) unknowns A_x and A_y can't be found until B is known. Inserting the known values into (3) and solving for B gives

$$\begin{aligned}
 B &= \frac{4(250) + 500}{6.098} \\
 &= \frac{1500 \text{ ft}\cdot\text{lb}}{6.098 \text{ ft}} \\
 &= 246.0 \text{ lb}
 \end{aligned}$$

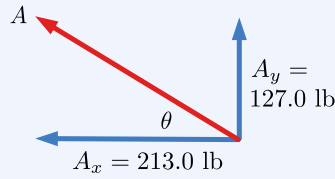
Now with the magnitude of B known, A_x and A_y can be found with (1) and (2).

$$\begin{aligned}
 A_x &= B \sin 60^\circ \\
 &= 246.0 \sin 60^\circ \\
 &= 213.0 \text{ lb}
 \end{aligned}$$

$$\begin{aligned}
 A_y &= C - B \cos 60^\circ \\
 &= 250 - 246.0 \cos 60^\circ \\
 &= 127.0 \text{ lb}
 \end{aligned}$$

The positive signs on these values indicate that the directions assumed on the free-body diagram were correct.

The magnitude and direction of force $\mathbf{A}$ can be found from the scalar components A_x and A_y using a rectangular to polar conversion.



$$A = \sqrt{A_x^2 + A_y^2} = 248.0 \text{ lb}$$

$$\theta = \tan^{-1} \left| \frac{A_y}{A_x} \right| = 30.8^\circ$$

The final values for **A** and **B**, with angles measured counter-clockwise from the positive x axis are

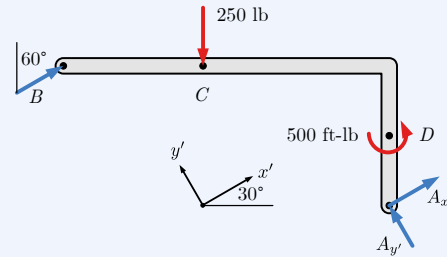
$$\mathbf{A} = 248.0 \text{ lb} \angle 149.2^\circ,$$

$$\mathbf{B} = 246.0 \text{ lb} \angle 30^\circ.$$

This solution demonstrates a fairly standard approach appropriate for many statics problems which should be considered whenever the free-body diagram contains a frictionless pin. Start by taking moments there.

Solution 2.

In this solution, we have rotated the coordinate system 30° to align it with force **B** and also chosen the components of force **A** to align with the new coordinate system.



There is no particular advantage to this approach over the first one, but with two unknown forces aligned with the x' direction, $A_{y'}$ can be found directly after breaking force C into components.

$$\begin{aligned} \sum F_{x'} &= 0 \\ B - C_{x'} + A_{x'} &= 0 \\ A_{x'} &= -B + C \sin 30^\circ \end{aligned} \quad (1)$$

$$\begin{aligned} \sum F_{y'} &= 0 \\ -C_{y'} + A_{y'} &= 0 \\ A_{y'} &= C \cos 30^\circ \end{aligned} \quad (2)$$

$$\begin{aligned} \sum M_A &= 0 \\ -B_x(3) - B_y(7) + C(4) + D &= 0 \\ 3B \cos 60^\circ + 7B \sin 60^\circ &= 4C + D \\ B(3 \cos 60^\circ + 7 \sin 60^\circ) &= 4C + D \end{aligned}$$

$$B = \frac{4C + D}{7.56} \quad (3)$$

Solving equation (2) yields

$$A_{y'} = 216.5 \text{ lb.}$$

Solving equation (3) yields the same result as previously

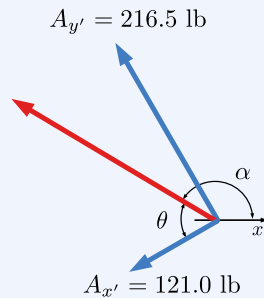
$$B = 246.0 \text{ lb.}$$

Substituting B and C into equation (1) yields

$$\begin{aligned} A_{x'} &= -B + C \sin 30^\circ \\ &= -246.0 + 250 \sin 30^\circ \\ &= -121.0 \text{ lb} \end{aligned}$$

The negative sign on this result indicates that our assumed direction for $A_{x'}$ was incorrect, and that force actually points 180° to the assumed direction.

Resolving the $A_{x'}$ and $A_{y'}$ gives the magnitude and direction of force $\mathbf{A}$.



$$A = \sqrt{A_{x'}^2 + A_{y'}^2} = 248.0 \text{ lb}$$

$$\theta = \tan^{-1} \left| \frac{A_{y'}}{A_{x'}} \right| = 60.8^\circ$$

$$\alpha = 180^\circ - (\theta - 30^\circ) = 149.2^\circ$$

Again, the final values for $\mathbf{A}$ and $\mathbf{B}$, with angles measured counter-clockwise from the positive x axis are

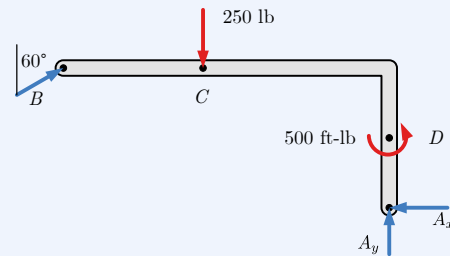
$$\mathbf{A} = 248.0 \text{ lb} \angle 149.2^\circ,$$

$$\mathbf{B} = 246.0 \text{ lb} \angle 30^\circ$$

This approach was slightly more difficult than solution one because of the additional trigonometry involved to find components in the rotated coordinate system.

Solution 3.

For this solution, we will use the same free-body diagram as solution one, but will use three moment equations, about points B , C and D .



$$\begin{aligned}
\sum M_B &= 0 \\
-A_x(3) + A_y(7) - C(3) + D &= 0 \\
-3A_x + 7A_y &= 250
\end{aligned} \tag{1}$$

$$\begin{aligned}
\sum M_C &= 0 \\
-A_x(3) + A_y(4) - B_y(3) + D &= 0 \\
-3A_x + 4A_y - 3B \cos 60^\circ &= -D \\
3A_x - 4A_y + 1.5B &= 500
\end{aligned} \tag{2}$$

$$\begin{aligned}
\sum M_D &= 0 \\
-A_x(1.5) - B_x(1.5) - B_y(7) + C(4) + D &= 0 \\
1.5A_x + 1.5B \sin 60^\circ + 7B \cos 60^\circ &= 4C + D \\
1.5A_x + 4.799B &= 1500
\end{aligned} \tag{3}$$

This set of three equations and three unknowns can be solved with some algebra.

Adding (1) and (2) gives

$$3A_y + 1.5B = 750 \tag{4}$$

Dividing equation (2) by 2 and subtracting it from (3) gives

$$2A_y + 4.049B = 1250 \tag{5}$$

Multiplying (4) by 2/3 and subtracting from (5) eliminates A_y and gives

$$3.049B = 750$$

$$B = 246.0 \text{ lb}$$

, the same result as before.

Substituting B into (3) gives $A_x = 213.0 \text{ lb}$, and substituting this into (1) gives $A_y = 127.0 \text{ lb}$, again the same result as before.

An alternate approach is to set these three equations up for a matrix solution and use technology to do the algebra, as done here with Sage.

$$\begin{bmatrix} -3 & 7 & 0 \\ 3 & -4 & 1.5 \\ 1.5 & 0 & 4.799 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ B \end{bmatrix} = \begin{bmatrix} 250 \\ 500 \\ 1500 \end{bmatrix}$$

```
A = Matrix([[ -3, 7, 0], [3, -4, 1.5], [1.5, 0, 4.799]])
B = vector([250, 500, 1500])
x = A.solve_right(B)
x
```

```
(213.020662512299, 127.008855362414, 245.982289275172)
```

This is a good example of an inefficient solution because of all the algebra involved. The issue here was the poor choice of B , C and D as moment centers. Whenever possible you should take moments about a point where the line of action of two unknowns intersect as was done in solution one. This gives a moment equation which can be solved immediately for the third unknown.

5.5 3D Rigid Body Equilibrium

Key Questions

- What are the similarities and differences between solving two-dimensional and three-dimensional equilibrium problems?
- Why are some three-dimensional reaction couple-moments “available but not engaged”?
- What kinds of problems are solvable using linear algebra?

Three-dimensional systems are closer to reality than two-dimensional systems and the basic principles to solving both are the same, however they are generally harder solve because of the additional degrees of freedom involved and the difficulty visualizing and determining distances, forces and moments in three dimensions.

Three-dimensional problems are usually solved using vector algebra rather than the scalar approach used in the last section. The main differences are that directions are described with unit vectors rather than with angles, and moments are determined using the vector cross product rather scalar methods. Because they have more possible unknowns it is harder to find efficient equations to solve by hand. A problem might involve solving a system of up to six equations and six unknowns, in which case it is best solved using linear algebra and technology.

Resolving Forces and Moments into Components. To break two-dimensional forces into components, you likely used right-triangle trigonometry, sine and cosine. However, three-dimensional forces will likely need to be broken into components using [Section 2.5](#).

When summing moments, make sure to consider both the $\mathbf{r} \times \mathbf{F}$ moments and also the couple-moments with the following guidance:

1. First, choose any point in the system to sum moments around.
2. There are two general methods for summing the $\mathbf{r} \times \mathbf{F}$ moments. Both techniques will give you the same set of equations.

(a) *Sum moments around each axis.*

For relatively simple systems with few position and force vector components, you can find the cross product for each non-parallel position and force pair. Using this method requires you to resolve the direction of each cross product pair using the right-hand rule as covered in [Chapter 4](#). Recall that there are up to six pairs of non-parallel components that you need to consider.

(b) *Sum all moments around a point using vector determinants.*

Choose a point in the system which is on the line of action of as many forces as possible, then set up each cross product as a determinant. After computing the components coming from each determinant, combine the x , y , and z terms into each of the $\Sigma \mathbf{M}_x = 0$, $\Sigma \mathbf{M}_y = 0$, and $\Sigma \mathbf{M}_z = 0$ equations.

3. Finally, add the components of any couple-moments into the corresponding $\Sigma \mathbf{M}_x = 0$, $\Sigma \mathbf{M}_y = 0$, and $\Sigma \mathbf{M}_z = 0$ equations.

Solving for unknown values in equilibrium equations. Once you have formulated $\Sigma \mathbf{F} = 0$ and $\Sigma \mathbf{M} = 0$ equations in each of the x , y and z directions, you could be facing up to six equations and six unknown values.

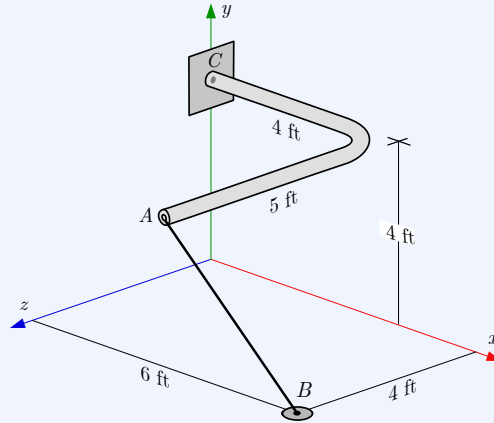
Frequently these simultaneous equation sets can be solved with substitution, but it is often be easier to solve large equation sets with linear algebra. Note that the adjective “linear” specifies that the unknown values must be linear terms, which means that each unknown variable cannot be raised to a exponent, be an exponent, or buried inside of a sin or cos function. Luckily, most unknowns in equilibrium are linear terms, except for unknown angles. If you are not familiar with the use of linear algebra matrices to solve simultaneously equations, search the internet for *Solving Systems of Equations Using Linear Algebra* and you will find plenty of resources.

No matter how you choose to solve for the unknown values, any numeric values which come out to be negative indicate that your initial hypothesis of that vector’s sense was incorrect.

Three-dimensional Equilibrium Examples.

Example 5.5.1 3D Bent Bar.

The bent bar shown is held in a horizontal plane by a fixed connection at C while cable AB exerts a 500 lb force on point A . Given $A = (4, 4, 5)$, $B = (6, 0, 4)$ and $C = (0, 4, 0)$.

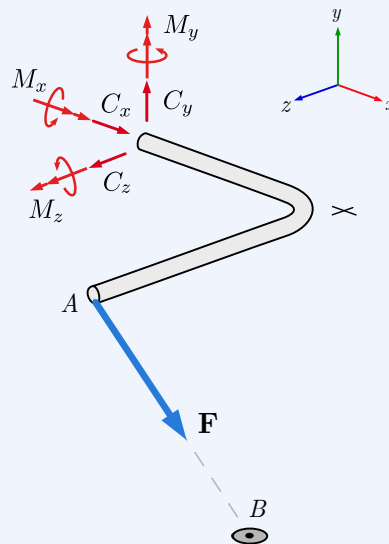


Find the reaction force $\mathbf{C}$ and concentrated moment $\mathbf{M}$ with components M_x , M_y and M_z required to hold the bar in this position under this condition.

Solution.

1. Draw a free-body diagram.

As always, begin by drawing a free-body diagram.



2. Determine the force acting at point A in Cartesian form.

The force of the cable acts from A to B . This direction is described

by the displacement vector from A to B

$$\mathbf{r}_{AB} = (2\mathbf{i} - 4\mathbf{j} - 1\mathbf{k}) \text{ ft}$$

or the corresponding unit vector

$$\begin{aligned}\lambda_{AB} &= \frac{\mathbf{r}_{AB}}{|\mathbf{r}_{AB}|} \\ &= \frac{2\mathbf{i} - 4\mathbf{j} - 1\mathbf{k}}{\sqrt{(2)^2 + (-4)^2 + (-1)^2}} \\ &= \frac{2\mathbf{i} - 4\mathbf{j} - 1\mathbf{k}}{\sqrt{21}}.\end{aligned}$$

Multiplying the unit vector by the cable tension gives the force acting on A as a three-dimensional Cartesian force vector.

$$\begin{aligned}\mathbf{F} &= \lambda_{AB}T \\ &= \left(\frac{2\mathbf{i} - 4\mathbf{j} - 1\mathbf{k}}{\sqrt{21}} \right) 500 \text{ lb} \\ &= (2\mathbf{i} - 4\mathbf{j} - 1\mathbf{k}) \left(\frac{500}{\sqrt{21}} \right) \text{ lb}\end{aligned}$$

$$\mathbf{F} = (218\mathbf{i} - 436\mathbf{j} - 109\mathbf{k}) \text{ lb}$$

3. *Determine the moment about C .*

The moment about point C is found with the [cross product \(4.3.4\)](#) where the moment arm is the displacement vector from C to A .

$$\mathbf{r}_{CA} = (4\mathbf{i} + 0\mathbf{j} + 5\mathbf{k}) \text{ ft}$$

$$\begin{aligned}\mathbf{M}_C &= \mathbf{r}_{CA} \times \mathbf{F} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 0 & 5 \\ 2 & -4 & -1 \end{vmatrix} \left(\frac{500}{\sqrt{21}} \right)\end{aligned}$$

$$\mathbf{M}_C = (2182\mathbf{i} + 1528\mathbf{j} - 1746\mathbf{k}) \text{ ft}\cdot\text{lb}$$

4. *Apply the equations of equilibrium.*

$$\Sigma \mathbf{F} = 0 \quad \left\{ \begin{array}{l} \Sigma F_x = 0 : \quad C_x + F_x = 0 \\ \qquad \qquad \qquad C_x = -218 \text{ lb} \\ \Sigma F_y = 0 : \quad C_y - F_y = 0 \\ \qquad \qquad \qquad C_y = +436 \text{ lb} \\ \Sigma F_z = 0 : \quad C_z - F_z = 0 \\ \qquad \qquad \qquad C_z = +109 \text{ lb} \end{array} \right.$$

$$\Sigma \mathbf{M} = 0 \quad \left\{ \begin{array}{l} \Sigma M_x = 0 : \quad M_x + M_{C_x} = 0 \\ \qquad \qquad \qquad M_x = -2182 \text{ ft}\cdot\text{lb} \\ \Sigma M_y = 0 : \quad M_y + M_{C_y} = 0 \\ \qquad \qquad \qquad M_y = -1528 \text{ ft}\cdot\text{lb} \\ \Sigma M_z = 0 : \quad M_z + M_{C_z} = 0 \\ \qquad \qquad \qquad M_z = +1746 \text{ ft}\cdot\text{lb} \end{array} \right.$$

The resulting vector equations for the reaction force $\mathbf{C}$ and reaction moment $\mathbf{M}$ are

$$\begin{aligned} \mathbf{C} &= (-218\mathbf{i} + 436\mathbf{j} + 109\mathbf{k}) \text{ lb} \\ \mathbf{M} &= (-2182\mathbf{i} - 1528\mathbf{j} + 1746\mathbf{k}) \text{ ft}\cdot\text{lb}. \end{aligned}$$

5.6 Stability and Determinacy

Key Questions

- What does *stable* mean for a rigid body?
- What does *determinate* mean for a rigid body?
- Does stability depend on the external loads or only on the reactions?
- How can I tell if a system is determinate?
- How can I decide if a problem is both stable and determinate, which makes it solvable statics?

Determinate vs. Indeterminate. A static system is **determinate** if it is possible to find the unknown reactions using the methods of statics, that is, by using equilibrium equations, otherwise it is **indeterminate**.

In order for a system to be determinate the number of unknown force and moment reaction components must be less than or equal to the number of independent equations of equilibrium available. Each equilibrium equation derives from a degree of freedom of the system, so there may be no more unknowns than degrees of freedom. This means that we can determine no more than three unknown reaction components in two-dimensional systems and no more than six in three-dimensional systems.

An indeterminate system with fewer reaction components than degrees of freedom is **under-constrained** and therefore unstable. On the other hand, if there are more reaction components than degrees of freedom, the system is both **over-constrained** and **indeterminate**. In terms of force and moment equations, there are more unknowns than equilibrium equations so they can't all be determined. This is not to say that it is impossible to find all reaction force on an over-constrained system, just that you will not learn how to find them in this course.

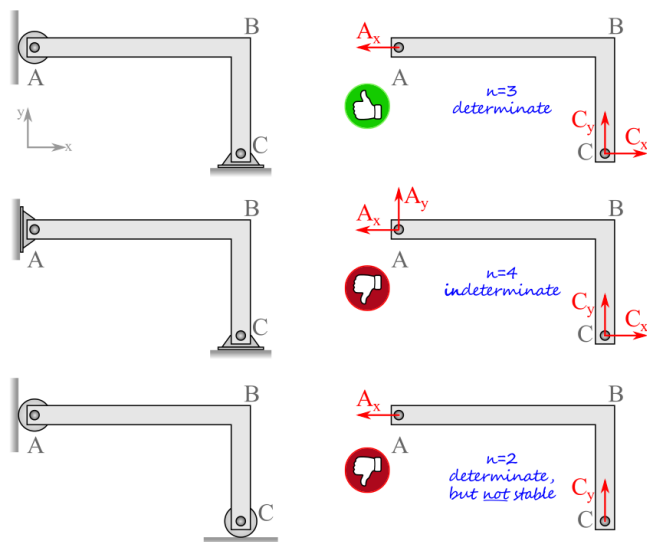
Stable vs. Unstable. A body in equilibrium is held in position by its supports, which restrict the body's motion and counteract the applied loads. When there are sufficient supports to restrain a body from moving, we say that the body is **stable**. A stable body is prevented from translating and rotating in all directions. A body which *can* move is **unstable** even if it is not currently moving, because the slightest change in load may take it out of equilibrium and initiate motion. Stability is loading independent i.e. a stable body is stable for *any* loading condition.

Rules to Validate a Stable and Determinate System. There are three rules to determine if a system is both stable and determinate. While, the rules below can technically be checked in any order, they have been sorted from the quickest to the most time consuming to speed up your analysis.

Rule 1: Are there exactly three reaction components on a two-dimensional body?

If YES, the system is determinate.

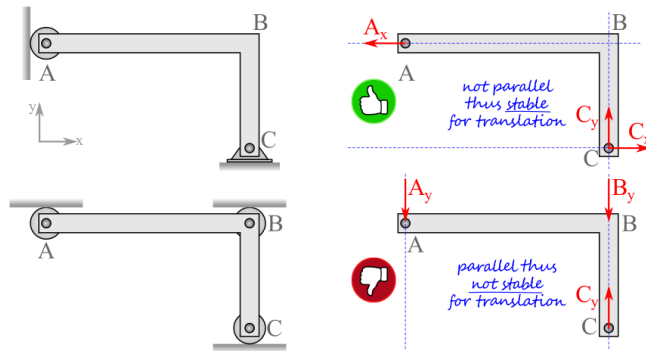
If NO, the system is indeterminate or not stable.



Rule 2: Are all the reaction force components parallel to one another?

If YES, the system is unstable for translation.

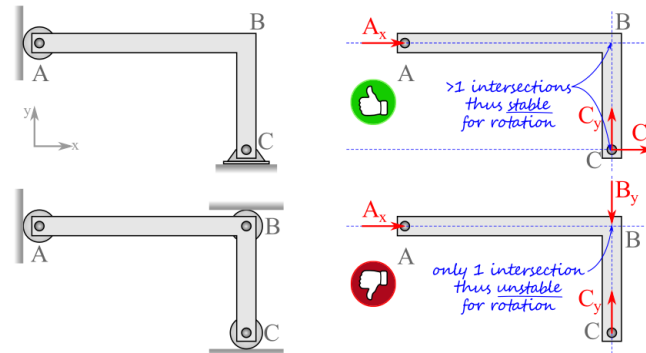
If NO, the system is stable for translation.



Rule 3: Do the lines of action of the reaction forces intersect at a single point?

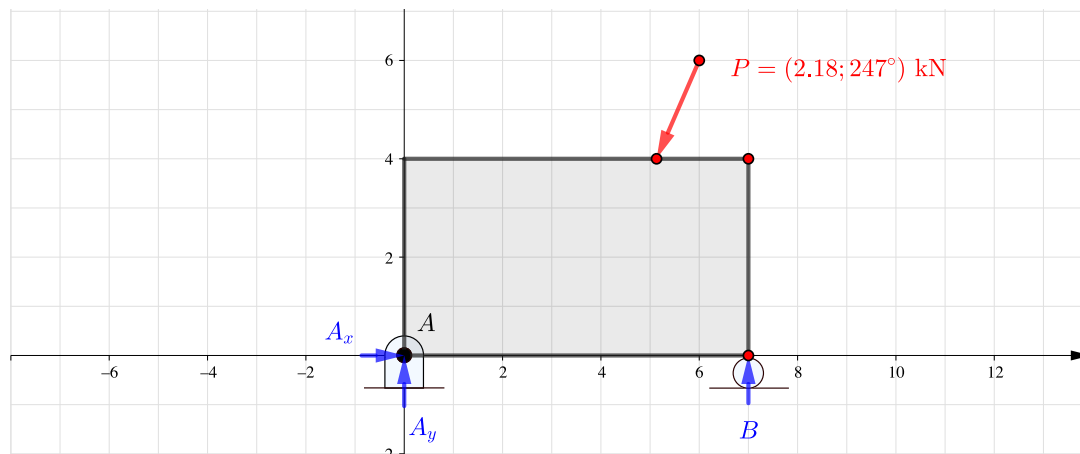
If YES, the system is unstable for rotation about the single intersection point.

If NO, the system is stable for rotation.



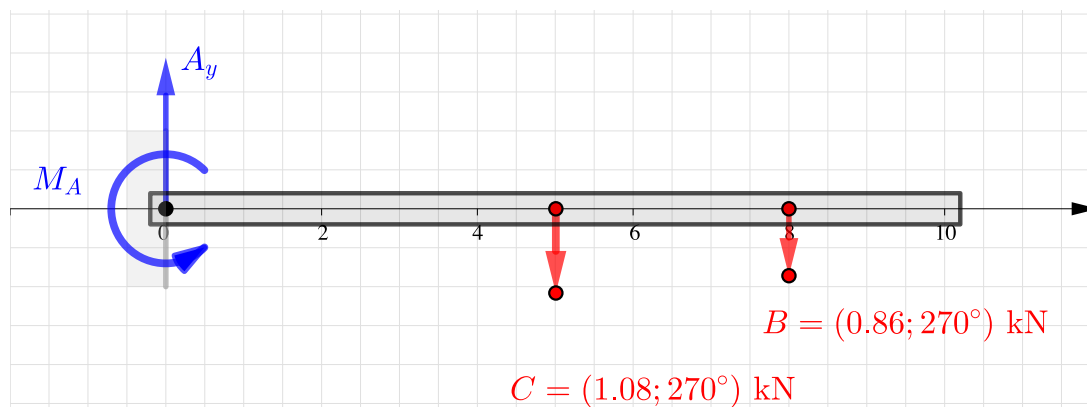
5.7 Equilibrium Examples

You can use these interactives to explore how the reactions supporting rigid bodies are affected by the loads applied. You can use the equations of equilibrium to solve for the unknown reactions, and check your work.



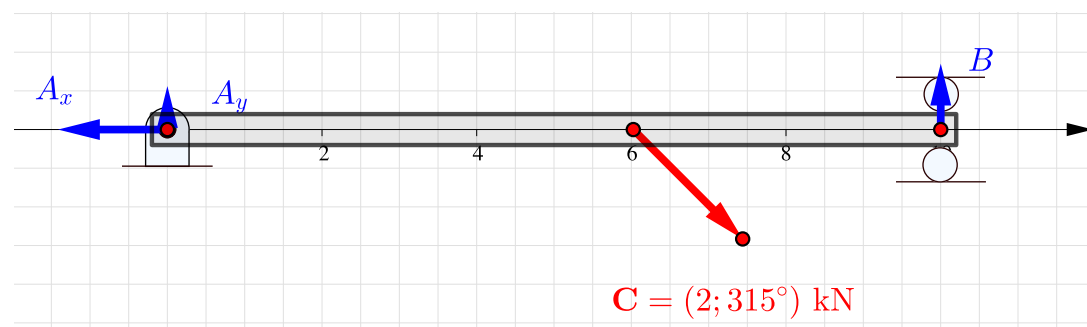
[Live Link](#)

Figure 5.7.1 Rigid body Equilibrium



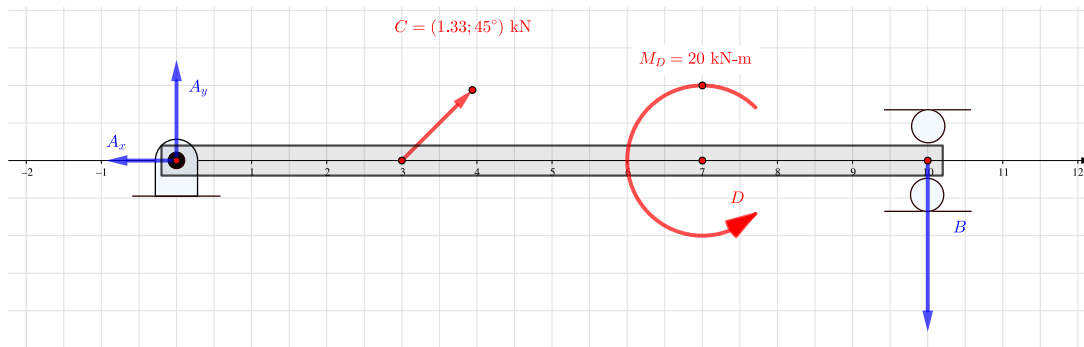
[Live Link](#)

Figure 5.7.2 Cantilever beam



[Live Link](#)

Figure 5.7.3 Beam with concentrated load



Live Link

Figure 5.7.4 Beam with concentrated force and couple moment

5.8 Exercises (Ch. 5)

Chapter 5 Exercises

Rigid Body Equilibrium		0/369
Tension in a Cable	0/4	Not attempted
Inclined Bar and cable	0/20	Not attempted
Isosceles Triangular Plate	0/30	Not attempted
Truss	0/60	Not attempted
Raise Pole	0/30	Not attempted
Triangle	0/50	Not attempted
Cantilever Beam	0/35	Not attempted
Beam and Pulley	0/40	Not attempted
Bell Crank	0/30	Not attempted
Car on a Hill	0/20	Not attempted
Hand Truck	0/20	Not attempted
Dump Truck	0/30	Not attempted
Three-force Body Method		0/55
Beam with Angled Load	0/20	Not attempted
Triangle	0/35	Not attempted

Display options

Done

Created using [Numbas](#) v10.0, developed by [Newcastle University](#)



Standalone
In Context